

Managerial Incentives and the Cost of Protectionism*

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Abstract

This paper shows that trade wars weaken managerial incentives, a key driver of firm productivity. It embeds a principal-agent problem in a two-country trade model with heterogeneous firms. Owners pay managers to reduce costs, and the optimal contract varies with the firm's initial productivity. Matched CEO-firm data for U.S. manufacturers support the model's predictions: incentive pay is hump-shaped in firm productivity, and its heterogeneous response to trade policy comes mainly from the tariffs U.S. exporters face abroad rather than from U.S. import tariffs. In the estimated model, a trade war strengthens incentives for less productive firms that sell only at home and weakens them for exporters, most strongly near the export margin. In the aggregate, incentives fall, and productivity and output decline by 6.5% and 5%, respectively. Models that omit managerial incentives miss about four fifths of the output cost of a trade war.

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JEL: F12, F13, F14, D24, M12, J33

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1 Introduction

The recent resurgence of protectionism has generated a large literature on the consequences of trade policy. These studies document the effects of tariffs on consumer prices (Amiti et al., 2019; Cavallo et al., 2021), firm employment and profitability (Flaaen and Pierce, 2024; Fajgelbaum et al., 2020), and aggregate welfare losses (Fajgelbaum et al., 2020). Most of this literature studies adjustment across firms: how tariffs reallocate market shares, affect participation in international markets, and induce firm entry and exit. However, we know much less about how firms adjust internally to changes in trade policy, or how these organizational responses affect productivity and aggregate output.

Managers exert costly effort that shapes firms’ productivity, and owners design incentive contracts to elicit that effort. A large body of evidence identifies management as a first-order determinant of productivity differences across firms and countries (Bloom and Van Reenen, 2007; Syverson, 2011; Bloom et al., 2013). Recent work further shows that trade reshapes managerial incentives, increasing executive pay and shifting compensation toward performance-based contracts (Hummels et al., 2026; Cuñat and Guadalupe, 2009). If trade policy rewrites managerial incentives, then protectionism affects not only which firms produce, but also how productively they operate.

This paper identifies managerial incentives as a new organizational margin through which trade policy affects productivity. We develop a heterogeneous-firm trade model in which the managerial incentive contract is an endogenous margin of adjustment to trade policy, derive its predictions for how incentives vary across firms, and test those predictions using matched CEO-firm data from U.S. manufacturers. We then estimate the model to quantify how changes in managerial incentives transmit the effects of a trade war to firm productivity and aggregate output.

The model formalizes a simple economic idea: the value of managerial effort depends on how much a firm sells. We embed a principal-agent problem in a heterogeneous-firm trade framework (Melitz and Ottaviano, 2008), in which owners design performance-based contracts to induce managers to exert costly effort that lowers production costs (e.g., by reorganizing operations, adopting new technologies, or improving efficiency). Because the returns to cost-reducing effort accrue on every unit sold, they increase with a firm’s effective market size. The optimal incentive contract therefore depends on two forces: market access and competition. Export barriers weaken managerial incentives directly, by reducing the foreign sales over which effort pays off. Import tariffs, by contrast, affect incentives only through their effect on competition in the domestic market.

This distinction leads to different responses across firms. Higher import tariffs soften domestic competition and increase the returns to cost-cutting effort, particularly for firms that sell primarily at home, although entry offsets part of this effect. Export barriers reduce the returns to effort among exporters by limiting their access to foreign markets. Since more productive firms are more likely to export, protectionism can strengthen incentives among smaller, domestic firms while weakening them among larger exporters. The costs of a trade war are consequently concentrated among exporters, particularly those closer to the export margin, while some smaller domestic firms may benefit.

To test these predictions, we combine matched CEO-firm data for publicly traded U.S. manufacturers from 1993 to 2007 with industry- and firm-level measures of U.S. import and export tariffs. We measure managerial incentives using CEO delta, the sensitivity of CEO wealth to firm value, a standard measure of performance-based incentives in the executive compensation literature (Core and Guay, 2002). Variation in import and export tariffs across firms, industries, and time allows us to identify how trade policy reshapes managerial incentives across the productivity distribution.

The data support the model's main predictions, confirming three distinct patterns. First, managerial incentives are hump-shaped in firm productivity. In the cross-section, incentive pay rises with productivity at first, peaks for intermediate-productivity firms, and then declines among the most productive firms. Second, the response of managerial incentives to trade policy is heterogeneous across the productivity distribution: a 10% reduction in export tariffs raises CEO delta by about 9% on average, although the response varies systematically with firm productivity. Third, the data display the asymmetry predicted by the model. While both import and export tariffs correlate with managerial incentives in levels, the heterogeneous responses across firms are driven by export tariffs, consistent with the model's market-access mechanism.

We then estimate the model using Simulated Method of Moments to quantify the importance of managerial incentives as a channel through which trade policy affects productivity. Two findings stand out. First, the productivity gains promised by import protection fail to materialize because the home import tariff has little direct effect on managerial incentives, while foreign retaliation, by contrast, weakens incentives and lowers both productivity and output. Second, managerial incentives are a first-order determinant of the output cost of a trade war. Output falls by five percent in the estimated economy with performance pay, but by only one percent in an otherwise identical economy without performance pay, implying that models without endogenous managerial incentives miss about four fifths of the output cost of a trade war.

This paper contributes to three strands of literature. Our primary contribution is to the literature on trade and protectionism. Existing work has emphasized how trade policy affects productivity through between-firm adjustment, including changes in selection, reallocation, and market shares (Melitz, 2003; Pavcnik, 2002; Trefler, 2004), as well as within-firm productivity improvements following trade liberalization, for example through technology upgrading (Bustos, 2011) and access to better intermediate inputs (Amiti and Konings, 2007). We identify a distinct organizational margin through which trade policy affects within-firm productivity by reshaping managerial incentives inside the firm. Quantitatively, this organizational margin is a first-order determinant of the output cost of a trade war.

Second, the paper contributes to the literature on management and productivity. A large body of evidence identifies management quality as an important source of productivity differences across firms and countries (Bloom and Van Reenen, 2007; Syverson, 2011; Bloom et al., 2013). This literature has focused mainly on ownership, competition, and informational frictions as determinants of managerial practices. This paper shows that trade policy is another important determinant of managerial incentives through changes in market access, with implications for firm productivity.

Finally, this paper contributes to the literature on managerial incentives. Existing theoretical work examines how competition shapes incentives in models with homogeneous firms (Hart, 1983; Schmidt, 1997; Raith, 2003), while recent empirical work shows that globalization and import competition affect executive compensation (Cuñat and Guadalupe, 2009; Hummels et al., 2026). This paper studies these forces in a heterogeneous-firm trade model with a principal-agent problem. Firm heterogeneity generates differences in how managerial incentives respond to trade policy across firms, which we use to quantify the implications for productivity and aggregate outcomes.¹

The rest of this paper is organized as follows. Section 2 presents the two-country model with managerial incentives, characterizes the optimal incentive contract and equilibrium, and derives the testable predictions for how trade policy reshapes managerial incentives across the productivity distribution. Section 3 describes the data. Section 4 presents the empirical strategy and evidence. Section 5 estimates the model and simulates the counterfactual trade-war experiments. Section 6 concludes.

¹Two related papers share some of these components but differ in important ways. Yu (2012) incorporates incentive contracts in a model with heterogeneous firms, but because managerial effort is chosen before the owner observes marginal cost, all firms end up with identical incentives. Chen (2014) incorporates a bargaining game in a Melitz (2003) model, but obtains unique predictions only for non-exporters; those predictions are consistent with ours.

2 The Model

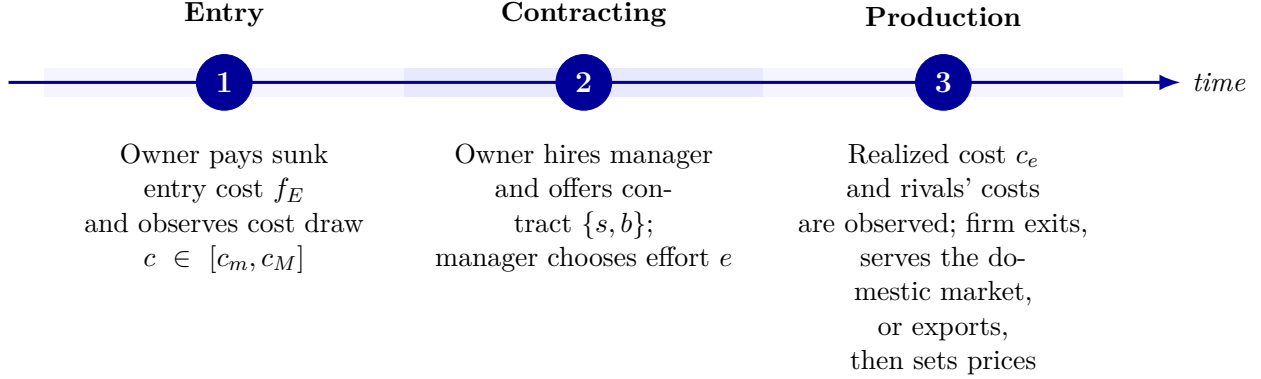
We extend the two-country model of monopolistic competition with heterogeneous firms developed by Melitz and Ottaviano (2008) by introducing a principal-agent problem between firm owners and managers. The economy consists of two countries, Home (H) and Foreign (F), with populations L_H and L_F . Countries share identical preferences and technologies but may differ in market size and trade costs. Firms produce differentiated varieties under monopolistic competition and are heterogeneous in productivity, operating under constant returns to scale with marginal cost $c = 1/\varphi$.

A key feature of the model is that the managerial incentive contract is determined endogenously. Owners design contracts to induce managers to exert unobservable, cost-reducing, effort. This creates an additional channel through which trade policy and competition affect firm productivity. In addition to the standard selection mechanism in heterogeneous-firm trade models, competition affects productivity by changing managerial incentives and, therefore, efficiency within firms. Because optimal incentives vary with firm productivity, firm heterogeneity determines the intensity of managerial effort across firms.

Timing. The model unfolds in three stages, illustrated in Figure 1. In the entry stage, a potential firm decides whether to incur the sunk entry cost f_E and, upon entry, observes its marginal-cost draw $c \in [c_m, c_M]$. In the contracting stage, the firm owner hires a manager in a competitive labor market,² offers an incentive contract, and the manager chooses an unobservable effort level. In the production stage, after observing the realized marginal cost c_e and the realized costs of competing firms, the firm decides whether to exit, serve only the domestic market, or serve both domestic and foreign markets, and sets prices in the markets it serves.

²Managers are assumed to be ex ante identical.

Figure 1: Time Structure of the Model



2.1 Consumption

Consumers have identical preferences in both countries, thus we suppress country subscripts to cause no confusion. Consumers are either production workers or managers. Production workers earn wage w_P , while managers receive compensation w_M and incur disutility from exerting effort. In equilibrium, managerial compensation offsets both the disutility of effort and the risk associated with the incentive contract, so workers and managers receive the same expected utility.

Following Melitz and Ottaviano (2008), preferences exhibit love of variety and generate variable-elasticity demand:

$$U = -\exp \left(-r \left[q_0^c + \alpha \int_{i \in \Omega} q_i^c, di - \frac{1}{2} \gamma \int_{i \in \Omega} (q_i^c)^2, di - \frac{1}{2} \eta \left(\int_{i \in \Omega} q_i^c, di \right)^2 - \psi(e) \right] \right), \quad (1)$$

where q_0^c and q_i^c denote per-capita consumption of the numeraire and differentiated goods, respectively, $q_i = Lq_i^c$ is market demand, Ω is the set of differentiated varieties, and $\psi(e)$ denotes the manager's disutility of effort. The preference parameters α , η , and γ are defined as in Melitz and Ottaviano (2008), while $r > 0$ is the coefficient of absolute risk aversion.

Market demand for variety i is

$$q_i = \frac{\alpha L}{\eta N + \gamma} - \frac{L}{\gamma} p_i + \frac{\eta N}{\eta N + \gamma} \frac{L}{\gamma} \bar{p}, \quad \forall i \in \Omega^*, \quad (2)$$

where $\Omega \subset \Omega$ denotes the set of consumed varieties, N is its measure, and $\bar{p}$ is the average

price across consumed varieties.³ Since demand is bounded, there exists a choke price p such that only varieties with $p_i < p$ are consumed:

$$p^* \equiv \frac{\gamma\alpha + \eta N \bar{p}}{\eta N + \gamma}. \quad (3)$$

The corresponding price elasticity of demand is $\varepsilon_i = \left[\frac{p^*}{p_i} - 1\right]^{-1}$, which increases with a firm's own price and decreases with the choke price. Following Melitz and Ottaviano (2008), we interpret p^* as a sufficient statistic for the intensity of competition: lower values of p^* correspond to more competitive markets.

The indirect utility associated with (2) is

$$\tilde{V}(p, I, e) = -\exp\left(-r\left(I - \psi(e) + \frac{1}{2}\left(\eta + \frac{\gamma}{N}\right)^{-1}(\alpha - \bar{p})^2 + \frac{1}{2}\frac{N}{\gamma}\sigma_p^2\right)\right),$$

where I denotes income and σ_p^2 is the cross-sectional variance of prices. This representation will be used to characterize the manager's optimization problem below.

2.2 Production and Firm Behavior

Labor is the only factor of production. Consumers supply labor inelastically in a competitive labor market. The numeraire good is produced competitively under constant returns to scale, implying a production-worker wage normalized to one, $w_P = 1$.

Potential firms incur a fixed entry cost f_E to enter the differentiated-goods sector. After entry, production takes place under constant returns to scale at realized marginal cost c_e , which depends on the outcome of the manager's project. Labor demand is therefore proportional to output,

$$l = c_e q.$$

Managers undertake a productivity-enhancing project whose outcome combines managerial effort and an idiosyncratic project shock according to

$$v = \delta e + u_i,$$

where $e \geq 0$ denotes managerial effort, $\delta > 0$ measures the effectiveness of effort, and u_i is an idiosyncratic project shock with mean zero.⁴ Greater managerial effort increases the expected

³ $\bar{p} = \int_{i \in \Omega} E[p_i], di.$

⁴For analytical convenience, we assume that u_i follows a truncated normal distribution on $[-\bar{u}, \bar{u}]$; the

success of the project, while the shock captures uncertainty beyond the manager's control. Managerial effort is therefore productive but not deterministic: it shifts the distribution of realized costs without eliminating project risk.

The project's outcome determines the firm's realized marginal cost,

$$c_e = g(c, v),$$

where $c \in [c_m, c_M]$, with $c_M \leq 1$, denotes the firm's initial marginal-cost draw observed after entry.

Assumption 1. *The realized marginal cost $c_e = g(c, v)$ is continuously differentiable in c and v and satisfies:*

$$(a) \ g'(c) > 0 \text{ and } g'(v) < 0$$

$$(b) \ g(c, v) \geq 0 \ \forall v \in [-\bar{u}, \bar{u}] \text{ and } c \in [c_m, c_M]$$

$$(c) \ \frac{\partial^2 g}{\partial c \partial v} < 0$$

Assumption A1 has a straightforward economic interpretation. Part (a) states that firms with higher initial marginal costs remain less efficient, while more successful productivity-enhancing projects reduce realized marginal costs. Part (b) guarantees that realized marginal costs are always non-negative. Part (c) implies that managerial effort is more effective in less productive firms: the marginal reduction in costs generated by additional effort is larger for firms with higher initial marginal costs.⁵

To obtain closed-form solutions, we specialize to the functional form

$$c_e = g(c, v) = c(1 - v),$$

so that realized marginal costs are

$$c_e = c(1 - \delta e - u_i),$$

and we restrict attention to effort levels such that $c_e \geq 0$ for all feasible realizations of u_i .

bound is firm-specific and is stated in Assumption A2(b) below.

⁵This assumption is consistent with the evidence in Bloom et al. (2013), who find that management practice interventions in Indian textile firms increased productivity and profitability, while similar interventions would have limited effects on firms already operating near best-practice levels.

The final component of the technology captures uncertainty in project outcomes. We assume that

$$\text{Var}(u_i) = \frac{\sigma^2}{c^2},$$

which implies

$$E[c_e] = c(1 - \delta e) \quad \text{and} \quad \text{Var}(c_e) = \sigma^2.$$

Thus, every productivity-enhancing project generates the same absolute variance, regardless of the firm's initial productivity. This assumption imposes a common level of uncertainty in project outcomes across firms and plays a central role in determining the optimal incentive contract. We discuss its implications after characterizing the contract below.

Markets are segmented, and firms may serve both domestic and foreign markets. Exporting from country l to country j requires iceberg trade costs $\tau_j > 1$, so that a firm with realized marginal cost c_e faces delivered marginal cost $\tau_j c_e$ abroad, while domestic sales are not subject to trade costs. Let $c_{l,D} = p_l^*$ denote the domestic survival cutoff in country l , and let $c_{l,X} = c_{j,D}/\tau_j$ denote the export cutoff faced by firms from country l . Since $\tau_j > 1$, only firms with sufficiently low realized marginal costs profitably export.

Conditional on the realized marginal cost c_e , firms maximize profits independently in each market they serve. The resulting optimal prices, quantities, and profits from country l are

$$p_{l,D}(c_e) = \frac{1}{2}[c_{l,D} + c_e] \quad q_{l,D}(c_e) = \frac{L_l}{2\gamma}[c_{l,D} - c_e] \quad (4)$$

$$p_{l,X}(c_e) = \frac{1}{2}[c_{l,X} + c_e] \quad q_{l,X}(c_e) = \frac{L_j}{2\gamma}\tau_j[c_{l,X} - c_e] \quad (5)$$

$$\pi_{l,D}(c_e) = \frac{L_l}{4\gamma}[c_{l,D} - c_e]^2 \quad \pi_{l,X}(c_e) = \frac{L_j}{4\gamma}\tau_j^2[c_{l,X} - c_e]^2. \quad (6)$$

Lower realized marginal costs translate into lower prices, higher sales, and greater profits in the domestic as well as the foreign market. Hence, firms with sufficiently low realized costs are both more likely to export and earn higher profits in foreign markets.

Owners compensate managers using linear incentive contracts,⁶ of the form

$$w_M = s + b(c_{l,D} - c_e),$$

⁶Although linear contracts are not generally optimal, they yield tractable closed-form solutions and are widely used in the literature on managerial incentives (see Bolton and Dewatripont (2005)). Extensions to non-linear contracts have been explored by Huang and Suarez (1997) and Hemmer et al. (2000).

where s is the fixed component of compensation and b is the performance-pay coefficient. The performance measure $(c_{l,D} - c_e)$ is a sufficient statistic for firm performance: it is proportional to firm output and to the markup, and is therefore observable and contractible. Equivalent results obtain if contracts are written on output or on the markup, which are proportional to $(c_{l,D} - c_e)$.

2.3 Manager's Optimization Problem

Given a compensation contract $\{s, b\}$, managers choose effort to maximize expected utility. Under CARA preferences, utility is

$$E_u[\tilde{V}(p, I, e)] = E_u[-\exp(-r(w_M - \psi(e) + Z))],$$

where

$$Z \equiv \frac{1}{2} \left(\eta + \frac{\gamma}{N} \right)^{-1} (\alpha - \bar{p})^2 + \frac{1}{2} \frac{N}{\gamma} \sigma_p^2$$

collects the consumer-surplus component of indirect utility that are independent of the manager's effort and compensation.

Following Chalioti (2014), with CARA preferences and a linear contract the manager's problem admits the mean-variance certainty-equivalent representation⁷

$$\max_e \left\{ E[w_M] - \frac{r}{2} \text{Var}(w_M) - \psi(e) \right\}.$$

We assume a quadratic effort cost,

$$\psi(e) = \frac{\theta e^2}{2},$$

where $\theta > 1$ measures the marginal cost of effort.⁸

The first-order condition implies

$$e^* = \frac{b\delta c}{\theta}. \tag{7}$$

Effort therefore increases with the strength of incentives (b), the effectiveness of managerial effort (δ), and the firm's initial marginal cost (c), and decreases with the marginal cost

⁷The representation is exact under normal noise. The bounded support of u_i adds a contract-dependent correction, derived in Online Appendix A. When we calibrate the model, it amounts to less than a third of one percent of the production wage for every firm, thus we work with the mean-variance representation throughout.

⁸The restriction $\theta > 1$ guarantees an interior solution.

of effort (θ).

Managers accept a contract whenever their expected utility is at least as large as their outside option. We assume the outside option is employment as a production worker, yielding utility

$$-\exp[-r(w_P + Z)].$$

The participation constraint is therefore

$$E_u[-\exp(-r(w_M - \psi(e) + Z))] \geq -\exp[-r(w_P + Z)].$$

Using the certainty-equivalent representation, the participation constraint becomes

$$E[w_M] - \frac{r}{2}\text{Var}(w_M) - \psi(e) \geq \bar{U}, \quad (8)$$

where

$$\bar{U} = w_P + \frac{\ln \Omega_i}{r}.$$

The term Ω_i reflects the truncation of the project shock. As $\bar{u} \rightarrow \infty$, the truncated distribution converges to the normal distribution, $\Omega_i \rightarrow 1$, and the reservation utility converges to w_P . The derivation is provided in Online Appendix A.

2.4 Firm's Optimization Problem

The firm owner chooses the compensation contract $\{s, b\}$ to maximize expected operating profits net of managerial compensation. Because the firm observes its realized marginal cost before choosing which markets to serve, expected operating profits account for the possibility that the firm exits, serves only the domestic market, or also exports. The owner's problem is

$$\begin{aligned} \max_{s,b} E_u[\Pi(c_e)] &= E_u[\pi_{l,D}(c_e) \mid c_e \leq c_{l,D}] P(c_e \leq c_{l,D}) \\ &\quad + E_u[\pi_{l,X}(c_e) \mid c_e \leq c_{l,X}] P(c_e \leq c_{l,X}) - E[w_M], \end{aligned} \quad (9)$$

subject to

$$\max_e E_u[w_M] - \frac{r}{2}\text{Var}(w_M) - \frac{\theta e^2}{2}, \quad (IC) \quad (10)$$

$$E_u[w_M] - \frac{r}{2}\text{Var}(w_M) - \frac{\theta e^2}{2} \geq \bar{U}, \quad (PC) \quad (11)$$

The concavity of the firm's objective and the manager's utility, together with CARA preferences and the distributional assumptions on u_i , ensures that the first-order approach is valid under the mean-variance representation. We therefore replace the incentive compatibility constraint (10) with the manager's optimal effort choice (7). Moreover, because the market for managers is competitive and frictionless, the participation constraint (11) binds in equilibrium. Solving the binding participation constraint yields the fixed component of compensation:

$$s = \bar{U} - \frac{b^2}{2\theta} [\delta^2 c^2 - \theta r \sigma^2] - b(c_{l,D} - c), \quad (12)$$

so that total compensation is

$$w_M = \bar{U} + \frac{b^2}{2\theta} [\delta^2 c^2 + \theta r \sigma^2] + cbu_i. \quad (13)$$

To guarantee the existence of an interior solution for the optimal incentive scheme, we impose the following assumptions:

Assumption 2. (a) $(\theta r \sigma^2 + \delta^2) \theta 2\gamma > (L_l + L_j \tau_j^2) \delta^4$

(b) $\bar{u}(c) = \frac{c_{l,D} - c}{c} \quad \forall c < c_{l,D}$

Assumption A2(a) guarantees concavity of profit in b . Assumption A2(b) makes the support of the project shock firm-specific: u_i is drawn from the normal distribution with variance σ^2/c^2 truncated to $[-\bar{u}(c), \bar{u}(c)]$, so the worst realization puts realized cost exactly at the survival cutoff ($c_e = c(1 + \bar{u}(c)) = c_{l,D}$ at zero effort). Realized costs therefore never exceed the cutoff, and domestic survival is certain for every entrant, the property used in the derivation of the optimal contract.⁹

Lemma 1. *Given Assumptions A2(a) and A2(b), the optimal incentive scheme for a firm in country l is*

$$b^*(c) = \frac{\delta^2 \theta [L_l(c_{l,D} - c) + L_j \tau_j^2(c_{l,X} - c)]}{2\theta \gamma (\theta r \frac{\sigma^2}{c^2} + \delta^2) - \delta^4 c^2 (L_l + L_j \tau_j^2)}, \quad (14)$$

⁹This assumption is standard in the financial economics literature; see Vives (1999, Ch. 8).

where the export term is present only for firms productive enough to export ($c < c_{i,X}$).

Proof: see Online Appendix B.¹⁰

The first term in the numerator, $L_i(c_{i,D} - c)$, captures the value of cost reductions in the domestic market. The second term, $L_j\tau_j^2(c_{i,X} - c)$, captures the value of cost reduction in export markets and is present only for exporting firms. Consequently, exporters face stronger incentive schemes than otherwise identical non-exporters. In the absence of export opportunities, $b^*(c)$ reduces to the closed-economy benchmark.

Substituting the optimal incentive scheme into the effort condition (7) yields the manager's optimal effort,

$$e^*(c) = \frac{\delta c}{\theta} b^*(c), \quad (15)$$

implying that managerial effort inherits the same cross-sectional pattern as incentives. Expected realized marginal costs are therefore

$$E[c_e] = c[1 - \delta e^*(c)],$$

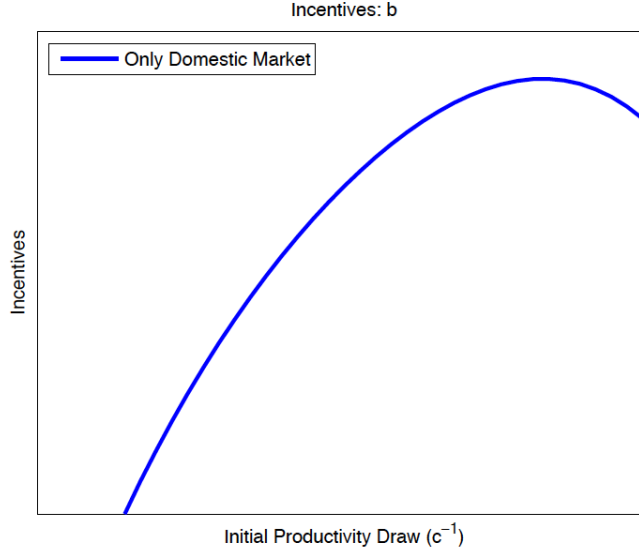
and are weakly lower than the initial cost draw c for all surviving firms.

Prediction 1 (Incentives and Productivity). *Managerial incentives are hump-shaped in firm productivity. The optimal bonus $b^*(c)$ is low among the least and most productive firms and reaches a maximum at an intermediate level of productivity.*

Figure 2 illustrates the equilibrium incentive schedule. Incentives are strongest among firms with intermediate productivity, where the gains from cost reductions are large enough to justify incentive provision but the associated compensation risk remains moderate.

¹⁰Under Assumption A2(b), every entrant survives domestically for all realizations of u_i , so the domestic term in the owner's problem is exact. The closed-form solution in Lemma 1 applies exactly to firms whose export status does not depend on realized costs. That is, firms that export under every realization and firms that never export. For firms near the export threshold, export status depends on the realized cost shock. The owner's optimal contract must therefore account for the probability that the firm exports. Section 5 solves this more general problem numerically. The closed-form and numerical incentive schedules coincide for firms outside the range around the export threshold.

Figure 2: Equilibrium Level of Incentives



The hump shape reflects a trade-off between incentive provision and risk sharing. As firms become more productive, they benefit more from reducing costs, giving owners a stronger incentive to reward managerial effort. At the same time, performance pay exposes managers to greater compensation risk, requiring firms to pay a larger risk premium. The benefits of stronger incentives dominate among less productive firms, while the cost of additional compensation risk dominates among the most productive firms.

The second force is where the variance specification $\text{Var}(u_i) = \sigma^2/c^2$ does its work: because every project puts the same absolute cost variance at stake, the risk premium per unit of initial cost rises with productivity, and it is this rising risk burden that eventually turns the bonus schedule back down. The assumption is therefore testable. Online Appendix C derives the optimal contract when $\text{Var}(u_i)$ is instead constant across firms and shows that the bonus would then be monotone in productivity. Prediction 1 thus distinguishes between the two specifications, and we take it to the data in Section 4.

2.5 Equilibrium with Free Entry

Entry is unrestricted in both countries, which share the same entry cost f_E and cost distribution $F(c)$. A firm in country l survives and produces if its expected profit is non-negative.

The marginal surviving firm, $\hat{c}_l$, satisfies the zero-cutoff-profit condition

$$E_u[\Pi(\hat{c}_l)] = E[\pi_{l,D}(\hat{c}_l) \mid c_e \leq c_{l,D}]P(c_e \leq c_{l,D}) + E[\pi_{l,X}(\hat{c}_l) \mid c_e \leq c_{l,X}]P(c_e \leq c_{l,X}) - E[w_M] = 0. \quad (16)$$

In the entry stage, firms enter until the expected value of entry equals the sunk entry cost:

$$\int_{c_m}^{\hat{c}_l} E_u[\Pi(c)] dF(c) = f_E. \quad (17)$$

Equations (16) and (17) for each country jointly determine the domestic cutoffs $\{c_{l,D}\}_{l=H,F}$ and survival cutoffs $\{\hat{c}_l\}_{l=H,F}$.

The zero-cutoff-profit and free-entry conditions imply that the domestic cutoff $c_{l,D}$ is decreasing in home-market size and increasing in trade costs. Larger home markets and lower trade costs raise the expected value of entry, attract additional entrants, and intensify competition, thereby lowering the domestic cutoff. The proof is provided in Online Appendix D.

The number of varieties sold in market l , denoted N_l , consists of domestic producers and foreign exporters:¹¹

$$F(c_{l,D})N_E^l + F(c_{j,X})N_E^j = N_l, \quad (18)$$

where N_E^l denotes the mass of entrants in country l . Lower cutoffs imply lower average realized costs and prices, thereby raising consumer welfare.

Labor markets clear in both countries. The corresponding condition, as well as the implied welfare expression, are reported in Online Appendix E.

A trade equilibrium is a collection $\{w_l, \hat{c}_l, c_{l,D}, N_E^l\}_{l=H,F}$ such that the zero-cutoff-profit condition (16), the free-entry condition (17), the measure-of-varieties condition (18), and labor-market clearing hold in both countries.

2.6 Trade Barriers and Managerial Incentives

We now characterize how trade policy reshapes managerial incentives. Trade barriers affect managerial incentives through two channels. The first is a *market-access effect*: trade barriers change the size of the market over which firms benefit from cost-reducing effort and, therefore, alter the returns to managerial incentives. The second is a *competition effect*: trade barriers affect firm entry and selection, shifting the survival cutoffs that determine the intensity of

¹¹Participation in these counts is indexed by the initial draw: exact for domestic survival under A2(b), while on the export margin the quantitative model determines status from realized cost.

competition. The market-access effect corresponds to the direct, partial-equilibrium effect of trade costs, while the competition effect captures the indirect, general-equilibrium response through entry and selection. Because managerial effort is proportional to the performance-pay coefficient given by equation (15), we focus on incentives throughout.

Consider first an increase in the trade cost faced by home firms exporting to the foreign market, τ_f . An increase in the export cost, τ_f , affects managerial incentives through two channels that work in opposite directions. The first is a market-access effect. A higher export cost lowers export revenue directly. Export receipts are the delivered price times the quantity sold abroad, $r_{H,X} = \tau_f p_{H,X} q_{H,X} = \frac{L_F}{4\gamma} (c_{F,D}^2 - \tau_f^2 c_e^2)$, so that

$$\frac{\partial r_{H,X}}{\partial \tau_f} = -\frac{L_F}{2\gamma} \tau_f c_e^2 < 0.$$

The loss scales with c_e^2 , so it is largest for exporters with high marginal costs, those closest to the export margin, and smaller for the most productive exporters. Lower export revenue reduces the return to cost-reducing effort. The market-access effect therefore weakens incentives, most strongly for firms with high marginal costs.

The second is a competition effect. A higher export cost removes some home exporters from the foreign market. Foreign firms face less competition, so the foreign survival cutoff $c_{F,D}$ rises. A higher foreign cutoff raises export revenue:

$$\frac{\partial r_{H,X}}{\partial c_{F,D}} = \frac{L_F}{2\gamma} c_{F,D} > 0,$$

and therefore strengthens incentives. However, in equilibrium, free entry dampens this effect: higher profits abroad attract foreign entrants, who push $c_{F,D}$ back down, so the net movement of the cutoff is determined in equilibrium. Export barriers therefore generate two opposing forces: a direct market-access effect that weakens exporters' incentives, most strongly near the export margin, and an indirect competition effect that strengthens them. Which force dominates is determined in equilibrium; Section 5 quantifies it.

Consider next an increase in the import barrier τ_h , the trade cost foreign firms face when selling in the home market. Import barriers differ from export costs in one key way: τ_h never enters the optimal incentive contract. Equation (14) depends on the domestic and foreign survival cutoffs and on the export cost τ_f , but not on τ_h . Import barriers therefore have no market-access effect. They move managerial incentives only through the competition effect, by shifting the equilibrium cutoffs.

The competition effect works first through the home market. A higher import barrier

removes some foreign exporters from the home market. Home firms face less competition, so on impact the domestic cutoff $c_{H,D}$ rises. Domestic revenue is $r_{H,D} = \frac{L_H}{4\gamma}(c_{H,D}^2 - c_e^2)$, so that

$$\frac{\partial r_{H,D}}{\partial c_{H,D}} = \frac{L_H}{2\gamma} c_{H,D} > 0.$$

A higher domestic cutoff raises revenue and strengthens incentives for all home firms. Free entry dampens this effect: higher profits at home attract new entrants, who push $c_{H,D}$ back down. Import barriers also lower foreign firms' export profits, which changes entry abroad and moves the foreign cutoff $c_{F,D}$. Through the export term of the contract, this in turn affects the incentives of home exporters. The overall effect of import barriers on incentives is therefore ambiguous, and its sign is determined in equilibrium.

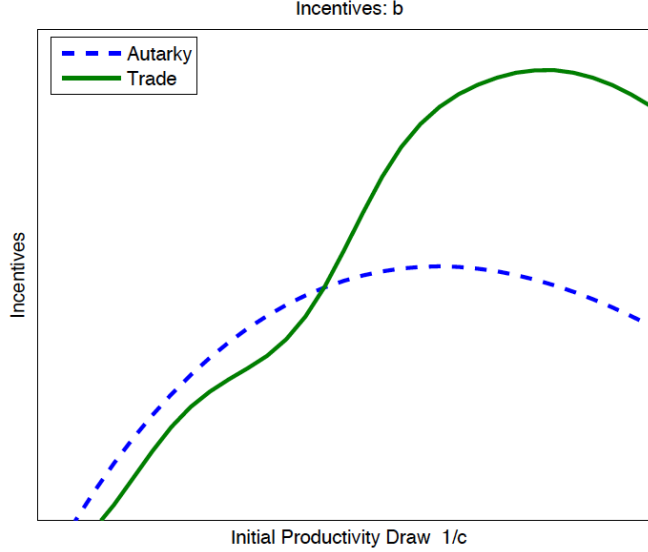
This highlights an important asymmetry in the model. Export costs affect incentives in two ways: directly, by limiting market access, and indirectly, by changing competition. Import barriers affect incentives in only one way: indirectly, through competition.

These effects are also heterogeneous across firms. As shown above, the market-access effect falls hardest on exporters near the export cutoff and much less on the most productive exporters. The competition effect, in contrast, works through the equilibrium cutoffs, so it reaches every firm that serves the affected market. Trade policy therefore reshapes managerial incentives unevenly across the productivity distribution rather than shifting them uniformly.

Prediction 2 (Trade Barriers and Managerial Incentives). *Trade barriers reshape managerial incentives heterogeneously across firms. Export barriers affect incentives both directly through market access and indirectly through competition, whereas import barriers affect incentives only indirectly through firm selection and competition.*

Figure 3 illustrates how trade changes the equilibrium incentive schedule. Relative to autarky, trade has the largest effects among firms in the middle and upper portions of the productivity distribution, while the least productive firms experience relatively small changes.

Figure 3: Incentives and Trade



Notes: The figure compares the equilibrium incentive schedule under autarky and trade. Parameter values are illustrative and are intended to convey mechanisms rather than magnitudes.

Together, Predictions 1 and 2 summarize the main empirical implications of the model. Prediction 1 describes the cross-sectional relationship between managerial incentives and firm productivity. Prediction 2 shows that the effects of trade-policy changes vary across the productivity distribution. In the empirical analysis, we test Prediction 1 using cross-sectional variation in firm productivity and Prediction 2 by examining whether firms at different points of the productivity distribution respond differently to tariff changes.

3 Data and Measurement

This section presents the primary datasets and variable construction.

3.1 Data

The paper combines three primary data sources covering the period 1993–2007. First, we collect firm- and CEO-level information from Compustat and CRSP. From Compustat, we use the Fundamentals Annual module to obtain firm financial data (sales, capital, employment, and materials), the Segments module to identify each firm’s lines of business at the 4-digit SIC level, and ExecuComp for CEO characteristics such as tenure. Stock prices and returns are obtained from CRSP. The sample is restricted to manufacturing firms (2-digit

SIC industries 20 through 39). To be included, firms must have complete information across these datasets over the sample period. Second, we obtain CEO incentive measures from the dataset compiled by Coles et al. (2006). This dataset reports CEO delta, a widely used measure of the sensitivity of CEO wealth to firm performance, constructed from ExecuComp following the methodology of Core and Guay (2002). Third, we complement these data with bilateral trade flows and tariff information from UNCTAD TRAINS.

3.2 Measurement

3.2.1 Productivity

We estimate total factor productivity at the firm level for our productivity measure. Assuming a Cobb-Douglas production function, our estimating equation takes the following form, in logs,

$$y_{it} = \beta_o + \beta_l l_{it} + \beta_k k_{it} + \beta_m m_{it} + \omega_{it} + \epsilon_{it}, \quad (19)$$

where y_{it} is the sales of firm i in period t deflated by the industry-level producer price index, l_{it} is the employment level measured as the number of employees, and k_{it} is the capital stock, measured as the gross property plant and equipment deflated by the non-residential private fixed investment,¹² and m_{it} is the expenditure in materials deflated by the material cost deflator.¹³ We estimate equation (19) following Akerberg et al. (2015) to overcome the well-known endogeneity problems. This procedure controls for the sample selection bias and the simultaneity bias between variable inputs and productivity shocks. We obtain separate coefficients for firms in each 2-digit SIC industry. Because sales are deflated by industry-level price indices, firm-specific prices and markups remain in the residual. We therefore interpret the estimate as revenue productivity rather than as a direct measure of technical efficiency. We use this measure as an empirical proxy for the firm’s position in the model’s productivity distribution

Using these estimates, we follow Pavcnik (2002) in constructing a firm-level productivity index that is transitive and insensitive to the units of measurement. The main goal is to be able to compare productivity measures across firms in different industries. We obtain such an index by subtracting from individual firm productivity the productivity of a reference

¹²As a robustness check, Online Appendix F re-estimates productivity using a perpetual-inventory capital stock constructed from gross property, plant, and equipment, capital expenditures, and depreciation. The results of Section 4 are unchanged.

¹³All the deflators come from the NBER-CES Manufacturing Industry Database.

firm in a base year,

$$\logprod_{it} = \log tfp_{it} - \log tfp_{93,I},$$

where $\log tfp_{it} = y_{it} - \hat{\beta}_l l_{it} - \hat{\beta}_k k_{it} - \hat{\beta}_m m_{it}$ and $\log tfp_{93,I}$ is the productivity of a reference firm for industry I. Reference firm is the firm with the average output and input in the year 1993, such that $\log tfp_{93,I} = \bar{y}_{it} - \hat{\beta}_l \bar{l}_{it} - \hat{\beta}_k \bar{k}_{it} - \hat{\beta}_m \bar{m}_{it}$, with the bar over the variable representing a mean over all firms in industry I for the year 1993. Because some specifications use the square of productivity, we shift the index by a positive constant so that it is positive for all firms (10 for the ACF-based index and 5 for the OLS-based index); the shifted variable is tfp_{it} . A common shift leaves firm rankings and the fitted quadratic relationship unaffected.

Panel A of Table 1 provides descriptive statistics for this index. Consistent with findings for other contexts, firm-level productivity distribution is dispersed and skewed to the right. Moreover, it highly correlates with sales, fixed assets, and market capitalization variables.

Table 1: Descriptive Statistics (1993-2007)

	Mean	SD	p25	p50	p75	N
<i>A. Firm Characteristics</i>						
Productivity Index (<i>tfp</i>)	10.37	1.01	9.88	10.14	10.49	9,512
Assets (US\$ M)	4059.36	14335.93	331.85	866.26	2599.50	10,170
Sales (US\$ M)	3811.85	12814.59	325.70	902.80	2572.51	10,170
Market Capitalization (US\$ M)	5908.01	20319.45	392.27	1003.14	3102.63	10,167
<i>B. CEO Characteristics</i>						
Tenure	8.04	7.36	3.00	6.00	10.00	10,171
Delta (US\$ M)	0.66	3.39	0.06	0.16	0.44	10,171
Holdings (%)	2.62	6.21	0.10	0.37	1.64	10,170
Cash Compensation (US\$ M)	1.13	1.01	0.52	0.83	1.38	10,171

Notes: Panel A shows descriptive statistics of main firm-level characteristics. Panel B corresponds to the CEO characteristics and features of the compensation scheme. The number of observations varies due to missing values on the variables.

3.2.2 Managerial Incentives

Our measure of managerial incentives is the CEO delta constructed by Coles et al. (2006). CEO delta measures the sensitivity of CEO wealth to firm performance and is defined as the dollar change in the value of the CEO's equity portfolio associated with a one-percent increase in the firm's stock price. This measure captures the strength of pay-for-performance incentives embedded in the CEO compensation package. The measure incorporates both stock and option holdings. Following Core and Guay (2002), the value of option holdings is estimated using information on exercisable and unexercisable options reported in ExecuComp

together with a modified Black–Scholes valuation formula. Because the dataset already provides the resulting delta measure, we do not recompute it and instead use the values reported by Coles et al. (2006). All dollar values are expressed in constant 2003 dollars. Panel B of Table 1 reports descriptive statistics for CEO delta. On average, a one-percent increase in the firm’s stock price raises CEO wealth by approximately \$660,000, while the median effect is about \$160,000. The distribution of CEO delta is highly right-skewed. We do not winsorize or trim the data; instead, our baseline specifications use the logarithm of CEO delta.

3.2.3 Trade Policy Measures

To measure trade policy, we construct export and import tariff measures using data from UNCTAD TRAINS. Export tariffs correspond to tariffs imposed on U.S. exports by foreign countries, while import tariffs correspond to tariffs imposed by the United States on foreign imports. Both measures are constructed at the industry and firm levels.¹⁴

At the industry level, we calculate export tariffs as the weighted average of the tariffs that foreign countries impose on U.S. exports within each 4-digit SIC industry. The weights are given by each country’s share in U.S. exports within a 4-digit SIC industry. Import tariffs are constructed analogously using U.S. tariffs on imports. At the firm level, tariffs are aggregated across the industries in which a firm operates using information from the Compustat Segments database. The weights are based on the share of firm sales accounted for by each 4-digit SIC industry.

To mitigate endogeneity concerns arising from changes in firms’ industrial composition or trade patterns, all tariff measures use fixed base-year weights from 1993. Identification therefore comes from variation in tariffs across industries, firms, and time rather than from changes in the weighting scheme.

Figure 4 plots average tariffs over the sample period. Export tariffs display substantially more variation than import tariffs. One important source of variation is the implementation of NAFTA. Prior to its entry into force in 1994, U.S. tariffs on Mexican imports averaged approximately 2.1%, whereas Mexican tariffs on U.S. exports averaged roughly 12%.

¹⁴The foreign countries considered are the ten largest U.S. trading partners, which together account for more than 80% of total U.S. trade.



Figure 4: Import and Export Tariff for the United States, 1993-2007

Notes: Figure shows the average firm-level tariffs per year. The solid line refers to import tariffs, while the dashed line corresponds to export tariffs.

Online Appendix Table 10 reports average tariffs by two-digit SIC industry. Tariff levels vary considerably across industries. Protection is higher in relatively labor-intensive industries, such as apparel and leather products, and lower in more capital-intensive industries, such as petroleum and paper. Tariffs declined over the sample period, but the pace of liberalization varied substantially across industries.

4 Testing the Model

This section tests the model's two main predictions. First, estimating the cross-sectional relationship between CEO incentives and firm productivity, we check whether managerial incentives show an inverted-U shape. Second, we check whether tariff changes move incentives differentially across the productivity distribution and whether export or import tariffs drive that movement.

4.1 Incentives and Productivity

We first examine the relationship between incentives and firms' productivity. To determine this relationship, we consider the following specification,

$$Incentives_{ijkt} = \beta_0 + f(tp_{j,k,93}) + \gamma_1 Z_{jt} + \gamma_2 Z_{kt} + \gamma_3 Z_{it} + \delta_i + \delta_k + \delta_t + \epsilon_{ijkt}, \quad (20)$$

where $Incentives_{ijkt}$ represent the incentives for the CEO i , working at firm j in the industry k at time t , measured by the CEO delta, and $f(tfp_{j,k,93})$ denotes a non-parametric expression to recover how incentives depend on firm-level productivity. We use two specifications. First, we use a quadratic specification for productivity. Then, we consider quintiles of base productivity. Industry level controls Z_{kt} are defined as follows. Industry entry cost refers to entry costs in a 4-digit SIC industry given by the natural log of the weighted average of the gross value of the cost of property, plant, and equipment, weighted by each firm’s market share. Industry market size is defined as the log sales at the four-digit SIC code level. Industry markup is the weighted average of a firm’s sales divided by its cost of goods sold, weighted by its market share. We also include firm-level controls Z_{jt} , such as the log of fixed assets and market capitalization. Additionally, in Z_{it} , we include the cash compensation associated with the payment scheme of the CEO. Finally, we control for executive (δ_i), industry (δ_k), and year (δ_t) fixed effects.

To alleviate the endogeneity concern of the effort induced by CEO’s incentives also determining firm productivity, we estimate equation (20) using the firm-level productivity measure at a base year, 1993. The underlying assumption is that baseline productivity has been determined before the CEO incentives in posterior years. We believe this assumption is conservative, as it might not be the case that firm-level productivity in 1993 matters for CEO incentives in 2007, making it more difficult for us to find an effect. We also re-estimate using OLS total factor productivity and lagged $(t - 2)$ productivity as robustness checks, reported in Online Appendix H; all the results hold in these cases. The two designs are complementary: baseline productivity has the advantage of being predetermined, while lagged productivity varies within executive-firm spells, so identification draws on the full panel; the inverse-U is reproduced with high significance (Columns 7–9 of Online Appendix Table 11).

The results are shown in Table 2. Columns 1 to 3 refer to the quadratic specification in productivity. Columns 4 to 6 correspond to the specification with quintiles. All specifications show a non-monotonic relationship between baseline productivity and incentives, consistent with the model predictions. In the quadratic specifications, the coefficient on baseline productivity is positive, suggesting a monotone relationship, while the coefficient of baseline productivity square is negative, indicating a non-monotonic, inverse-U relationship. This non-monotonic relationship holds for specifications that only consider fixed effects (Column 1), after adding industry-level controls (Column 2), and even when controlling for CEO and firm-level characteristics (Column 3). The implied turning point is stable across specifications (between 9.5 and 9.7, against a median baseline productivity of 10.0) and lies between the 10th and 25th percentiles of the distribution, so the sample contains firms on

both sides of the peak. Moreover, consistent with the model and previous literature (Raith, 2003; Karuna, 2007), an increase in market size or a decrease in industry entry cost both raise managerial incentives. In addition, wealthier firms tend to expand managerial incentives.

These results are confirmed under a more flexible specification that lets the productivity-incentive slope differ across baseline-productivity quintiles (Columns 4 to 6). The estimated slopes are largest for firms in the lower quintiles and smallest, and statistically insignificant in the full specification, for the fourth quintile (the top quintile is the omitted reference). That is, managerial incentives rise steeply with productivity among less productive firms but flatten toward the top of the distribution, consistent with the non-monotonic relationship in Figure 2. Beyond confirming Prediction 1, the inverse-U also discriminates between variance specifications of the model: with a constant project-shock variance across firms, the optimal bonus would be monotone in productivity (Online Appendix C), so the non-monotonic pattern in the data supports the baseline specification.

	Quadratic Specification			Quintiles		
	(1)	(2)	(3)	(4)	(5)	(6)
$\log \omega_{j93}$	21.200*** (3.683)	22.859*** (3.416)	8.938* (4.808)			
$\log \omega_{j93}^2$	-1.099*** (0.190)	-1.177*** (0.177)	-0.471* (0.256)			
q1 - $\log \omega_{j93}$				1.883** (0.782)	1.803** (0.777)	1.576*** (0.494)
q2 - $\log \omega_{j93}$				1.688*** (0.523)	1.573*** (0.508)	1.270*** (0.348)
q3 - $\log \omega_{j93}$				1.390** (0.691)	1.417** (0.688)	1.368*** (0.307)
q4 - $\log \omega_{j93}$				0.595 (0.386)	0.677* (0.368)	0.271 (0.223)
Controls	No	Yes	Yes	No	Yes	Yes
R-Squared	0.89	0.89	0.91	0.89	0.89	0.91
N. Obs	6278	6278	5630	6278	6278	5630

Table 2: Productivity and Incentives (CEO Delta)

Notes: The dependent variable is the log CEO delta. Columns 1–3 use a quadratic in baseline (1993) log productivity ($\log \omega_{j93}$); Columns 4–6 replace it with productivity-quintile slopes (q1–q4, top quintile omitted). All columns include executive, industry (2-digit SIC), and year fixed effects. Columns 2 and 5 add industry controls (segment entry cost, sales, markup); Columns 3 and 6 add firm and CEO controls (assets, market capitalization, cash compensation). Standard errors clustered by firm in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% levels.

4.2 Trade Liberalization and Incentives

We now turn to examine the impact of tariff reductions on incentives. We use the following equation:

$$Incentives_{ijkt} = \beta_0 + \beta_1 f(\tau_{jt}, tfp_{j,k,93}) + \gamma_1 Z_{kt} + \gamma_2 Z_{jt} + \gamma_3 Z_{it} + \delta_k + \delta_i + \epsilon_{ijkt} \quad (21)$$

where τ_{jt} are firm-level export (or import) tariffs imposed to (by) the U.S. by (to) its main trading partners, constructed by aggregating industry tariffs across each firm's segments with fixed 1993 sales weights (as described in section 3.2.3); $tfp_{j,k,93}$ is baseline (1993) productivity as in equation (20); Z_{kt} , Z_{jt} , and Z_{it} are industry, firm, and executive variables as described previously; and δ_i and δ_k refer to executive and industry fixed-effects, respectively. Unlike equation (20), this specification does not include year fixed effects: the time variation in tariffs is central to identification, and year effects would absorb the economy-wide decline in tariffs over the sample period. The aggregate tariff coefficients below should be read accordingly, as they are identified in part from that common decline. The heterogeneous responses across productivity quintiles, which are the model's sharp prediction, are robust to including year fixed effects (Online Appendix I).

The model's sharp prediction is not about the average effect of a tariff change but about its heterogeneity: trade-policy changes reshape the incentive-productivity relationship differentially across firms, whereas a homogeneous-firm model implies a uniform effect. To examine this, we first assess the impact of export and import tariffs alone, and then interact them with the firm's baseline productivity quintiles.

Results are reported in Table 3. The first three columns show the aggregate effect of export and import tariffs, controlling only for the battery of fixed effects mentioned previously. Column 1 shows the estimates of incentives as a function of export tariffs. Tariffs enter as $\log(1 + \tau)$, with τ the ad-valorem rate in percent, so the coefficient of -0.92 implies that a fall in the export tariff from a 5.0% to a 4.5% ad-valorem rate (a 10% relative fall) increases CEO incentives by about 8.0%, a mean increase in CEO delta of roughly \$53,000. A similar estimate is found in Column 2 when considering import tariffs: the same relative fall increases CEO incentives by about 7.3%, roughly \$48,000 at the mean. When both tariffs are included simultaneously (Column 3), both coefficients remain statistically significant, though the export coefficient is roughly twice as large as the import coefficient.

To see whether the decrease in export tariffs affects firms differently depending on their initial productivity, we look at the interaction terms in equation (21). These are shown in

Columns 4 to 6. All specifications include all the industry-, firm-, and executive-level controls and fixed effects. Column 4 shows the interactions of our baseline productivity measure only with export tariffs, Column 5 considers the exact specification only with import tariffs, and Column 6 includes both.

The results corroborate the model’s heterogeneity prediction. In the full specification (Column 6), the export-tariff interaction with baseline productivity is significant and varies systematically across the distribution: the top-quintile response itself is a precisely estimated zero (0.001, standard error 0.074), so the interaction coefficients coincide with the quintiles’ total responses, which decline monotonically in magnitude from -0.69 in the lowest quintile to essentially zero in the highest.

The model also predicts that export and import tariffs have different impacts on incentives. In particular, because the import tariff never enters the firm’s incentive contract, the heterogeneous incentive response across firms should mostly run through export tariffs, while import tariffs correlate with incentives only indirectly through general-equilibrium selection. This is a demanding test: import and export tariffs are highly correlated over this period¹⁵, so each coefficient could easily absorb part of the other’s variation. In levels, both tariffs are important. When both are included (Column 3), they remain significant, with the export coefficient roughly twice the import coefficient, consistent with the indirect, selection channel the model allows for import tariffs. The sharp evidence is in the interactions. When import tariffs enter alone (Column 5), three of their four interactions with productivity appear significant; once export tariffs are added (Column 6), only one import–productivity interaction remains significant, while the export–productivity interactions are significant and decline monotonically across quintiles whether import tariffs are included or not (Columns 4 and 6). This is the pattern one expects if the export tariff is the true driver of the heterogeneous response and the import interactions in Column 5 were proxying for it. The heterogeneity of the incentive response is thus identified from export tariffs, the core of the contract mechanism.

As robustness checks, we re-estimate equation (21) with a quadratic tariff-productivity interaction and with OLS total factor productivity quintiles in place of the baseline ACF quintiles; the results, reported in Online Appendix J, are consistent with the baseline, with the export-tariff response remaining heterogeneous across the productivity distribution.

¹⁵A simple OLS regression between log of export tariffs and log of import tariffs leads to a coefficient of 0.74, significant at the 1% level

	Tariffs			Interaction		
	(1)	(2)	(3)	(4)	(5)	(6)
Export Tariff	-0.920*** (0.051)		-0.742*** (0.055)	-0.007 (0.073)		0.001 (0.074)
Import Tariff		-0.842*** (0.090)	-0.398*** (0.074)		-0.029 (0.075)	-0.074 (0.079)
q1 - $\log\omega_{j93}$				2.637*** (0.638)	1.996*** (0.681)	2.716*** (0.707)
q2 - $\log\omega_{j93}$				1.585*** (0.527)	1.260** (0.504)	1.644*** (0.530)
q3 - $\log\omega_{j93}$				2.216*** (0.494)	1.855*** (0.461)	2.327*** (0.501)
q4 - $\log\omega_{j93}$				0.939** (0.476)	0.787* (0.456)	1.084** (0.481)
Export Tariff * q1 - $\log\omega_{j93}$				-0.663*** (0.209)		-0.686*** (0.214)
Export Tariff * q2 - $\log\omega_{j93}$				-0.526*** (0.134)		-0.390*** (0.131)
Export Tariff * q3 - $\log\omega_{j93}$				-0.348*** (0.126)		-0.295** (0.125)
Export Tariff * q4 - $\log\omega_{j93}$				-0.326*** (0.120)		-0.240** (0.118)
Import Tariff * q1 - $\log\omega_{j93}$					-0.366 (0.295)	0.062 (0.306)
Import Tariff * q2 - $\log\omega_{j93}$					-0.589*** (0.141)	-0.337*** (0.129)
Import Tariff * q3 - $\log\omega_{j93}$					-0.357** (0.154)	-0.167 (0.150)
Import Tariff * q4 - $\log\omega_{j93}$					-0.332** (0.142)	-0.176 (0.144)
Controls	No	No	No	Yes	Yes	Yes
R-Squared	0.83	0.82	0.84	0.89	0.89	0.89
N. Obs	8520	8520	8520	5630	5630	5630

Table 3: Trade Shock and Incentives (CEO delta)

Notes: The dependent variable is the log CEO delta. Columns 1–3 report the aggregate effect of export and import tariffs (Column 1, export; Column 2, import; Column 3, both), including executive and industry (2-digit SIC) fixed effects only. Columns 4–6 interact export and import tariffs with baseline (1993) productivity quintiles (q1–q4, top quintile omitted) and add the full set of industry, firm, and CEO controls (the “Controls” row). Standard errors clustered by firm in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% levels.

5 Quantifying the Cost of Protectionism

This section takes the model to the data to answer two questions: what are the effects of a trade war in an economy where firms use performance pay to sustain productivity, and by how much would we understate or overstate those effects if we ignored the performance-pay channel? We proceed in two steps. The first half of this section estimates the model’s structural parameters and discusses identification. The second half uses the estimated economy to perform two counterfactuals. First, we simulate a trade war to understand the cost of protectionism. Second, to gauge the importance of the organizational margin, we compare this case to an economy without performance pay.

5.1 Simulated Method of Moments

We estimate seven structural parameters using the Simulated Method of Moments (SMM): the entry cost f_E , the moral-hazard parameters δ (marginal product of effort), θ (effort disutility), and σ (cost noise), the Pareto shape κ , and the bilateral trade costs τ_f (on home exports) and τ_h (on home imports). We fix the product-differentiation parameter at $\gamma = 0.25$.¹⁶ The estimation targets eight moments: seven from 1993 firm-level data on revenues, CEO incentives, and export status pin down the shape of the model, and an effort-contribution anchor from the management-and-productivity literature pins down the scale of the moral-hazard block.

Externally set parameters. We treat country market sizes as observed rather than estimated. The home economy is the United States and the foreign economy is the rest of the world. We set $L = 1$ and $L_f = 3$, reflecting that the U.S. accounts for roughly one quarter of world GDP (a ratio of 1:3 relative to the rest of the world).¹⁷ We further set the coefficient of absolute risk aversion to $r = 2$, the cost-draw support to $[c_m, c_M] = [0.05, 0.9]$, and the manager’s outside option $\bar{U}$ to the average employee wage.¹⁸

¹⁶The parameter γ primarily scales revenues and bonuses and is weakly identified by our moments; we discuss robustness to this choice in Online Appendix L.

¹⁷We discuss robustness to this assumption in Online Appendix K.

¹⁸Average annual earnings of stable employees in 2000, \$36,699, from the Census Bureau’s Quarterly Workforce Indicators, expressed in model units (millions of dollars).

Estimator. The estimator minimizes the weighted distance between model-simulated and empirical moments:

$$\hat{\Theta} = \arg \min_{\Theta} \left[m(\Theta) - m^{data} \right]' W \left[m(\Theta) - m^{data} \right], \quad (22)$$

where W is diagonal. Each of the seven data moments enters with $W_{jj} = 1/|m_j^{data}|$, so it contributes its squared deviation divided by the absolute value of its empirical target. The deviation of the effort-contribution anchor is scaled by an additional factor of 30, so that in practice the anchor binds and the estimated model matches it exactly.

Parameter search. We evaluate the criterion on a 10,000-point Halton grid over the seven-dimensional parameter space: at each candidate we solve for the equilibrium cutoffs $(c_{H,D}, c_{F,D})$, discard candidates without a stable equilibrium or with a non-concave bonus schedule (Assumption A2a), and evaluate the moments on 1,000 simulated firms.

5.2 Identification

Although the parameters are determined jointly by the full set of targets, each target is most informative about a particular parameter. We describe this mapping below.

The role of γ . Product differentiation γ multiplies the slope of demand, so it primarily scales revenues, quantities, and bonus levels rather than their distribution across firms. It is not, however, a pure scale parameter: γ also enters the compensation-risk term of the optimal contract, so we do not claim exact invariance. In practice γ is weakly identified by our moments: when we estimated it jointly with the other parameters, the criterion was nearly flat in γ and the estimate drifted across otherwise similar specifications. We therefore fix $\gamma = 0.25$ and choose moments that do not load on the levels γ primarily governs: the median-demeaned log-revenue spreads $p_{25} - p_{50}$ and $p_{75} - p_{50}$, the scale-free bonus interquartile ratio p_{75}/p_{25} , the markup, and the bonus-productivity regression coefficients. Online Appendix L re-estimates the model at $\gamma = 0.15$ and $\gamma = 0.35$: the criterion is nearly flat, the anchored effort contribution is unchanged, and the counterfactual conclusions are essentially unaffected.

Selection and the cost distribution. The entry cost f_E and the Pareto shape κ govern selection and dispersion. In equilibrium, free entry equates the expected value of entry to f_E , so a higher entry cost moves the equilibrium cutoffs and the mass of survivors. Thus, f_E

is identified jointly by the export participation share and the markup, which discipline the equilibrium cutoffs. The shape κ controls the dispersion of cost draws (a smaller κ implies a fatter right tail and greater dispersion) and is identified by the log-revenue spreads.

Trade costs. The export cost τ_f acts directly on home firms: it sets the export threshold $c_{F,D}/\tau_f$ and scales export revenue, and is identified by the export participation share together with the markup. The import cost τ_h does not enter home firms' problem directly; it affects home outcomes only in general equilibrium, by changing the intensity of import competition and hence the domestic cutoff. It is identified by the markup, which summarizes the competitiveness of the domestic environment.

Managerial incentives: shape. The shape of the bonus schedule is identified by the bonus moments. The noise parameter σ governs the compensation-risk term $\theta r \sigma^2 / c^2$ in the optimal contract; this term rises as productivity rises and is what bends the bonus schedule into its inverse-U shape. We exploit this curvature by targeting the coefficients of the auxiliary regression

$$\log b^*(c) = \beta_0 + \beta_1 \cdot \frac{1}{c} + \beta_2 \cdot \left(\frac{1}{c}\right)^2, \quad (23)$$

where $1/c$ denotes firm productivity, together with the scale-free bonus interquartile ratio. The linear coefficient $\beta_1 > 0$ captures the increasing portion of the incentive schedule and the quadratic coefficient $\beta_2 < 0$ captures its concavity. The empirical counterparts of β_1 and β_2 are taken from the lagged-productivity specification (Columns 7–9 of Online Appendix Table 11), which draws on the full panel.

Managerial incentives: scale. The shape moments pin down only combinations of the moral-hazard parameters (δ, θ, σ) : every targeted bonus moment is invariant to the scale of the bonus, so the criterion is nearly flat along a ridge on which the implied managerial effort varies widely (Online Appendix M). We resolve the ridge by anchoring the contribution of managerial effort to within-firm productivity, the productivity loss from switching performance pay off at fixed cutoffs, at 6%, within the range of a one-standard-deviation improvement in management practices (Bloom and Van Reenen, 2007). The effort scale is thus disciplined by this anchor choice. However, Online Appendix N re-estimates the model with the anchor at the extremes of the evidence range, 3% and 8%. The magnitudes scale with the anchor, but the qualitative conclusions are unchanged.

5.3 Results

Table 4 reports the parameter estimates and model fit. All estimates are interior to the parameter space.

The estimated marginal product of effort is $\delta = 1.72$ and the effort disutility $\theta = 9.53$. The noise standard deviation $\sigma = 0.21$ implies substantial project risk relative to the estimated survival cutoff. The Pareto shape $\kappa = 2.28$ implies wide dispersion in initial cost draws, within the range of estimates in the trade literature. The estimated trade costs, $\tau_f = 1.29$ and $\tau_h = 1.86$, imply moderate barriers in both directions, higher on imports than on exports. At these estimates the equilibrium cutoffs are $c_{H,D} = 0.57$ and $c_{F,D} = 0.50$.

The model fits the moments well. The bonus interquartile ratio is matched almost exactly (5.36 vs. 5.44 in the data), as well as the two bonus-productivity coefficients ($\beta_1 = 1.01$ vs. 1.01; $\beta_2 = -0.059$ vs. -0.046). Thus, the model reproduces the inverse-U incentive relationship that motivates the empirical analysis quite well. The model underestimates the markup a bit (1.29 vs. 1.48), and the log-revenue spreads are compressed relative to the data. The latter is a common feature of Melitz-type models, which generate dispersion only through the productivity draw.

Online Appendix Table 17 reports the standard errors. The noise parameter σ and the export cost τ_f are precisely estimated, the Pareto shape less sharply, and the import cost τ_h is imprecise, which is consistent with its indirect, general-equilibrium identification. As mentioned before, δ and θ are individually imprecise, as what is well identified is their combination. The anchor instead pins the effort contribution itself tightly.¹⁹

5.4 Counterfactuals

We run two counterfactuals, holding the structural parameters fixed and re-solving the equilibrium. The first evaluates the cost of protectionism by raising the import trade cost, the export trade cost, and both together. The second quantifies how much of the change caused by protectionism comes from the incentive channel.²⁰

¹⁹Standard errors are computed by the delta method from the numerical moment Jacobian, a firm-level bootstrap of the empirical moments, and the simulation-noise covariance. Also, since the effort-contribution anchor is a normalization, they are themselves conditional on it.

²⁰The counterfactuals below are accordingly conditional on the anchored effort scale; Online Appendices M and K report their sensitivity across θ and across foreign-market size.

Table 4: SMM Estimation Results

<i>Panel A: Parameter Estimates</i>		
Parameter	Estimate	
Entry cost (f_E)	2.77×10^{-2}	
Marginal product of effort (δ)	1.715	
Noise std. dev. (σ)	0.209	
Pareto shape (κ)	2.275	
Effort disutility (θ)	9.528	
Product differentiation (γ , fixed)	0.250	
Trade cost, home $\rightarrow$ foreign (τ_f)	1.287	
Trade cost, foreign $\rightarrow$ home (τ_h)	1.864	
<i>Panel B: Model Fit</i>		
Moment	Data	Model
p25–p50 log revenue	−0.983	−0.734
p75–p50 log revenue	1.177	0.864
Bonus p75/p25 ratio	5.438	5.363
Export share	0.464	0.535
Markup (sales/COGS)	1.482	1.295
Bonus-productivity: linear	1.012	1.011
Bonus-productivity: quadratic	−0.046	−0.059

Notes: Seven parameters ($f_E, \delta, \sigma, \kappa, \theta, \tau_h, \tau_f$) estimated by simulated method of moments (Halton grid search refined by local optimization), matching the seven moments in Panel B together with an effort-contribution anchor that disciplines the scale of the moral-hazard block (see text; minimized criterion 0.19). Product differentiation γ is fixed at 0.25 (see text). Log-revenue moments are matched as median-demeaned spreads. The equilibrium cutoffs are $c_D = 0.573$ (domestic) and $c_D^f = 0.497$ (foreign). Bonus-productivity coefficients are from a regression of log bonus on firm productivity ($1/c$) and its square. Standard errors are reported in Online Appendix Table 17.

5.5 Protectionist Trade Policy

We simulate the cost of a trade war like the one initiated under the second Trump administration. The first scenario is the home country (the United States) raising its own import tariff (τ_h). Then, we consider the scenario where trading partners raise the barrier home exporters face abroad (τ_f). We conclude with the two together, simulating a full trade war. Each trade-cost wedge rises by 0.18 from its estimated baseline, an increase comparable to the rise in the U.S. effective tariff rate observed in 2025.²¹ Table 5 reports the three scenarios in that order. We report the survival rate, export participation, incentive pay, output, productivity, and the cost cutoffs that drive selection. The survival rate refers to the share of a fixed pool of potential entrants that remain active in equilibrium. Similarly, the exporter share is the share of active firms that export, with export status determined by each firm’s realized cost. Incentive pay is summarized by the mean bonus paid across firms, and output is aggregate sales (total firm revenue). Productivity, measured as revenue-weighted inverse average realized cost $1/\bar{c}_e$, is our welfare-relevant summary. All effects are shown keeping the cutoffs fixed and in full equilibrium, which adds entry and selection. All effects are reported as percentage changes relative to each economy’s own baseline.

Table 5: Counterfactual: Simulating a Trade War (trade costs +0.18)

Outcome	Import τ_h		Export τ_f		Bilateral	
	Fixed cutoffs	Full eq.	Fixed cutoffs	Full eq.	Fixed cutoffs	Full eq.
Survival rate (%)	+0.0	-1.6	-0.3	+14.2	-0.3	+12.6
Exporter share (%)	+0.0	+2.8	-21.7	-33.7	-21.7	-29.7
Mean bonus	+0.0	+2.4	-23.5	-19.0	-23.5	-18.6
Output	+0.0	+0.2	-13.9	-5.1	-13.9	-5.0
Productivity $1/\bar{c}_e$	+0.0	+0.8	-0.1	-7.6	-0.1	-6.5
Home cutoff $c_{H,D}$	0.573	0.568	0.573	0.612	0.573	0.607
Foreign cutoff $c_{F,D}$	0.497	0.501	0.497	0.491	0.497	0.496

Notes: Each scenario raises the indicated trade-cost wedge by 0.18 from its estimated baseline. Entries are percentage changes relative to baseline, except the cutoffs $c_{H,D}$ and $c_{F,D}$, reported as levels (the fixed-cutoff column shows the baseline level at which they are held). Baseline levels: survival rate 31.0%, exporter share 55.8%, mean bonus 0.157. The exporter share here is computed from a different realization of the cost noise than the estimation moment in Table 4 and therefore differs slightly.

Import protection does little. The case commonly made for import tariffs, and the one advanced in recent U.S. trade policy, is that shielding domestic firms from foreign competition will revitalize domestic industry and productivity. The counterfactual does not support that

²¹<https://budgetlab.yale.edu/research/state-us-tariffs-july-28-2025>

claim. A higher home import tariff (τ_h) has no significant effect on the managerial channel. This is precisely because of its indirect nature. The tariff τ_h does not appear in the firm's incentive schedule, so at a fixed cutoff, bonuses, effort, output, and productivity are all unchanged (the entire fixed-cutoff column is zero). Import protection does not lead firms to become more productive. It does operate through selection. However, even there the effect is faint. By lowering foreign competition at home, it raises domestic profits, drawing in marginally more entry and thus tightening the cutoff slightly (from 0.573 to 0.568). As a result, productivity increases by 0.8% and a slightly larger share of surviving firms export (+2.8%). This is a small, second-order gain that operates through selection, yet it is not due to an improvement in how firms are organized.

Lost export access is costly. Let us now consider a higher barrier on home exports to a foreign country (τ_f). This impacts productivity through two reinforcing mechanisms. The first is our novel managerial channel. A firm's optimal bonus rewards cost reduction in proportion to the sales over which a lower cost pays off, and the export component of that schedule scales with foreign-market size and declines with the export barrier. When τ_f rises, the foreign-market return to cost reduction shrinks, so firms scale back the incentives they offer (mean bonuses fall by about 24%), managers exert less effort, and realized costs rise. Export participation contracts sharply (the exporter share falls by 22%, partly because weaker effort itself pushes marginal firms out of exporting) and output falls by 14%. Measured productivity, however, barely moves at fixed cutoffs (-0.1%). This is because weaker effort raises the costs of continuing firms, but it also pushes the weakest producers out of the market, and the two effects on average offset each other. Thus, the losses through the managerial channel are concentrated mostly on output and pay. The second mechanism is selection, and it is this one that drives the productivity loss. Lower export profits depress the value of entry, so fewer firms enter, competition at home weakens, and the survival cutoff rises from 0.573 to 0.612. Less productive firms that the baseline would have screened out now survive (the survival rate rises by about 14%), and aggregate productivity falls by 7.6%. Taken together, a loss of export-market access erodes both the incentive to become productive and the selection that keeps out the unproductive.

A full trade war is a net loss. The bilateral experiment, in which both barriers rise together, lowers productivity by 6.5%. The small selection gain from import protection is overwhelmed many times over by the losses from reduced export access, which operate through both weaker managerial incentives and weaker selection. The protectionist policy

of import tariffs does not contribute to productivity or output enhancement, whereas the retaliation it provokes creates substantial losses. The managerial-incentive channel we identify is a new channel from trade policy to productivity, distinct from the between-firm selection mechanism of Melitz (2003) or Melitz and Ottaviano (2008); while quantitatively smaller than the latter, it is still economically significant.

5.6 The Role of Managerial Bonuses

To gauge the importance of the incentive contract, we perform a counterfactual where we remove the managerial project altogether: without performance pay ($b = 0$ for all firms) managers exert no effort, and with no project there is no project risk, so firms produce at their drawn cost. Results are shown in Table 6. The effect on baseline productivity is large: the no-bonus economy is 14.0% less productive than the estimated economy. Both margins explain this. At a fixed cutoff, shutting down the managerial project already lowers productivity by 6% and cuts export participation by about 27%, as effort no longer carries marginal firms across the export threshold. Survival at fixed cutoffs actually rises (by 15%), because removing the project also removes its noise, sparing marginal firms that a bad realization would have pushed out. General equilibrium, then, more than doubles the loss. Without the profits that incentive pay helps generate, fewer firms are able to cover the cost of entry. Therefore, the home cutoff rises from 0.573 to 0.621 and much less productive firms remain active. Managerial incentives are therefore a first-order determinant of aggregate productivity: removing them costs more than twice as much productivity as the trade war itself (14.0% versus 6.5%).

Table 6: Counterfactual: Removing Managerial Bonuses

Outcome	Fixed cutoffs	Full eq.
Survival rate (%)	+15.2	+37.7
Exporter share (%)	-26.7	-32.4
Mean bonus	-100.0	-100.0
Productivity $1/\bar{c}_e$	-6.0	-14.0
Home cutoff $c_{H,D}$	0.573	0.621
Foreign cutoff $c_{F,D}$	0.497	0.520

Notes: The no-bonus economy removes the managerial project: $b = 0$ for all firms, so there is no effort and no project noise, and firms produce at their drawn cost. Entries are percentage changes relative to baseline (the fixed-cutoff columns hold the cutoff at its baseline; the full-equilibrium columns re-solve it), except the cost cutoffs $c_{H,D}$ and $c_{F,D}$, which are reported as levels. Productivity is revenue-weighted $1/\bar{c}_e$. For the cutoffs, the fixed-cutoff column reports the baseline level at which they are held. Baseline levels: survival rate 31.0%, exporter share 55.8%, mean bonus 0.157.

The incentive mechanism also changes what a trade war costs. In an economy where firms use bonuses to raise productivity, protectionism inflicts a loss that standard models miss: tariffs compress the instrument firms use to elicit effort. To quantify it, we re-run the trade-war scenarios in the no-bonus economy and compare the responses. They are each measured in percentage terms against that economy’s own baseline (a comparison that, unlike output levels, is invariant to the demand normalization). Table 7 reports the results. For aggregate productivity the two economies suffer similarly, since that cost runs through selection, which operates in both economies. For output the difference is stark. With performance pay, a bilateral war reduces output by 5.0% and export retaliation by 5.1%; without it, by only 1.0% and 1.2%. The mechanism is the one at the center of the model: when export access shrinks, firms serve a smaller market and thus cut the bonuses they offer. Managers then decrease effort, which in turn cuts output. By contrast, an economy without performance pay has no effort to withdraw. Tariffs are thus costly here for the standard selection reasons, and the managerial channel makes them substantially worse: about four fifths of the output cost of a trade war runs through it; thus, a model that omits managerial incentives understates that cost by a factor of four to five. This share is robust to the anchored effort scale: with the anchor at the conservative end of the evidence range, the channel still accounts for roughly two thirds of the output cost, and the understatement factor remains at least three (Online Appendix N).

Table 7: The Trade War With and Without Performance Pay

Scenario	With performance pay		Without performance pay	
	Productivity	Output	Productivity	Output
Import τ_h	+0.8	+0.2	+0.7	+0.2
Export τ_f	-7.6	-5.1	-9.0	-1.2
Bilateral	-6.5	-5.0	-8.3	-1.0

Notes: General-equilibrium effect of each tariff scenario on revenue-weighted productivity ($1/\bar{c}_e$) and output, in the estimated economy and in the counterfactual economy with performance pay switched off ($b = 0$ for all firms). Each scenario raises the indicated barrier by 0.18. All entries are percentage changes relative to each economy’s own baseline, a comparison that is invariant to the demand normalization.

6 Conclusion

Standard accounts of protectionism operate between firms, through selection and reallocation. This paper shows that trade policy also operates inside the firm, through the managerial-incentive contract. Embedding a principal-agent problem into a two-country

Melitz-Ottaviano model, we show that the optimal contract is priced on market access: incentives are hump-shaped in firm productivity, export barriers enter the contract directly, and the import tariff never does. Matched CEO-firm data for U.S. manufacturers confirm these predictions: the CEO delta is non-monotonic in productivity, and its heterogeneous response to trade policy is identified from export rather than import tariffs, robust to OLS and lagged productivity measures.

Estimating the model and simulating a trade war of the kind initiated under the second Trump administration yields a clear policy message. Import protection is essentially incentive-neutral and does almost nothing for productivity; the damage comes from the retaliation it provokes, so that a bilateral trade war leaves the country that raised tariffs with lower productivity and lower output than before the war. The managerial channel we identify is a new route from trade policy to productivity, distinct from between-firm selection: modest for aggregate productivity, but a first-order determinant of the level of productivity and the chief conduit through which a trade war depresses output. Extending the framework to dynamic or non-linear contracts, to endogenous innovation, and to settings outside the United States is left to future work.

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Online Appendix

A Derivation of the Manager's Reservation Wage

Under the contract $w_M = s + b(c_{l,D} - c_e)$ and the realized cost $c_e = c(1 - \delta e - u_i)$, the manager's compensation can be written as

$$w_M = s + b[c_{l,D} - c(1 - \delta e)] + b c u_i,$$

where the only random component is $b c u_i$. With constant absolute risk aversion, the manager's expected utility is

$$\begin{aligned} E_u[-\exp(-r(w_M - \psi(e) + Z))] &= -\exp\left(-r[s + b(c_{l,D} - c(1 - \delta e)) - \psi(e) + Z]\right) \\ &\quad \times E_u[\exp(-r b c u_i)]. \end{aligned}$$

Taking logs and dividing by $-r$, the certainty equivalent is

$$\text{CE} = s + b[c_{l,D} - c(1 - \delta e)] - \psi(e) + Z - \frac{1}{r} \log E_u[\exp(-r b c u_i)].$$

When u_i is normal, $-\frac{1}{r} \log E_u[\exp(-r b c u_i)] = -\frac{r}{2} b^2 c^2 \text{Var}(u_i) = -\frac{r}{2} b^2 \sigma^2$, the standard mean-variance penalty, where the last equality uses $\text{Var}(u_i) = \sigma^2/c^2$. Since in our case u_i is a *symmetric truncated* normal on $[-\bar{u}, \bar{u}]$, the moment-generating function carries an additional factor equal to the ratio of truncation probabilities evaluated at the tilted mean. When we collect this correction in the term Ω_i , the binding participation constraint $\text{CE} = w_P + Z$ can be written as in equation (8) with the effective reservation value

$$\bar{U} = w_P + \frac{\ln \Omega_i}{r}, \quad \Omega_i = \frac{\Phi\left(\frac{\bar{u}c}{\sigma} + r b \sigma\right) - \Phi\left(\frac{-\bar{u}c}{\sigma} + r b \sigma\right)}{\Phi\left(\frac{\bar{u}c}{\sigma}\right) - \Phi\left(\frac{-\bar{u}c}{\sigma}\right)},$$

where Φ is the standard normal c.d.f. and $\bar{u}c/\sigma$ is the standardized support bound of u_i . Since $\Omega_i \leq 1$, the correction effectively decreases the reservation value. This is because the truncated noise has less risk than the normal one, so the manager requires less compensation to overcome it: $\bar{U} \leq w_P$. As the truncation is eliminated, $\bar{u} \rightarrow \infty$, the truncated noise distribution converges to the untruncated normal, $\Omega_i \rightarrow 1$, and the correction goes to zero.

Given that the truncation window is symmetric, the first-order effect of the tilt $rb\sigma$ cancels, therefore $\ln \Omega_i$ is of order $(rb\sigma)^2$. We check this statement in our calibration. At the estimated parameters, $|\ln \Omega_i|/r$ is below 0.33 percent of the production wage along the equilibrium bonus schedule, and below 0.11 percent at the mean bonus; treating $\bar{U}$ as fixed in the owner's problem is therefore innocuous.

B Proof of Lemma 1 (Optimal Incentive Scheme)

The owner chooses b to maximize expected operating profit net of expected compensation, replacing the incentive-compatibility constraint by the manager's optimal effort $e^* = b\delta c/\theta$ (the first-order approach is valid by the concavity established under Assumption A2). Throughout, we treat the mean-variance certainty equivalent as exact; Online Appendix A quantifies the truncation correction and shows it is negligible at the estimated parameters. Using $c_e = c(1 - \delta e - u_i)$ and $\text{Var}(u_i) = \sigma^2/c^2$,

$$E[\pi_{l,D}] = \frac{L_l}{4\gamma} E[(c_{l,D} - c_e)^2] = \frac{L_l}{4\gamma} \left\{ [c_{l,D} - c(1 - \delta e)]^2 + \sigma^2 \right\},$$

and analogously $E[\pi_{l,X}] = \frac{L_j\tau_j^2}{4\gamma} \left\{ [c_{l,X} - c(1 - \delta e)]^2 + \sigma^2 \right\}$ for exporters. Substituting $e^* = b\delta c/\theta$ gives $c(1 - \delta e) = c - \frac{\delta^2 c^2}{\theta} b$, hence

$$c_{l,m} - c(1 - \delta e) = (c_{l,m} - c) + \frac{\delta^2 c^2}{\theta} b, \quad m \in \{D, X\}.$$

From the binding participation constraint, expected compensation is $E[w_M] = \bar{U} + \frac{b^2}{2\theta} (\delta^2 c^2 + \theta r \sigma^2)$. Dropping terms independent of b , the owner's objective is

$$\Pi(b) = \frac{L_l}{4\gamma} \left[(c_{l,D} - c) + \frac{\delta^2 c^2}{\theta} b \right]^2 + \frac{L_j\tau_j^2}{4\gamma} \left[(c_{l,X} - c) + \frac{\delta^2 c^2}{\theta} b \right]^2 - \frac{b^2}{2\theta} (\delta^2 c^2 + \theta r \sigma^2).$$

The first-order condition $\Pi'(b) = 0$ is

$$\frac{\delta^2 c^2}{2\gamma\theta} \left[L_l(c_{l,D} - c) + L_j\tau_j^2(c_{l,X} - c) \right] + b \left[\frac{\delta^4 c^4}{2\gamma\theta^2} (L_l + L_j\tau_j^2) - \frac{\delta^2 c^2 + \theta r \sigma^2}{\theta} \right] = 0.$$

Solving for b and multiplying numerator and denominator by $2\gamma\theta/c^2$ yields

$$b^*(c) = \frac{\delta^2\theta[L_l(c_{l,D} - c) + L_j\tau_j^2(c_{l,X} - c)]}{2\theta\gamma(\theta r\sigma_u^2 + \delta^2) - \delta^4c^2(L_l + L_j\tau_j^2)},$$

which is equation (14). Assumption A2(a) guarantees that the denominator is positive, so $\Pi(b)$ is strictly concave and $b^*(c)$ is the unique maximizer. The export term is present only for firms productive enough to export, $c < c_{l,X}$. Otherwise the $L_j\tau_j^2$ terms drop out and $b^*(c)$ collapses to the closed-economy case.

C The Variance Specification and the Shape of the Incentive Schedule

The baseline model imposes $\text{Var}(u_i) = \sigma^2/c^2$. Thus, the variance of realized marginal costs, $\text{Var}(c_e) = \sigma^2$, is the same across firms. This specification is not innocuous, as it makes the compensation-risk term in the optimal contract, $\theta r\sigma^2/c^2$, rise with firm productivity, and that rising risk term is what leads to the inverse-U of Prediction 1.

What would the alternative be? Suppose instead that the noise variance is constant across firms, $\text{Var}(u_i) = \sigma_u^2$. The variance of compensation is then $\text{Var}(w_M) = b^2c^2\sigma_u^2$, expected compensation under the binding participation constraint is $E[w_M] = \bar{U} + \frac{b^2}{2\theta}(\delta^2c^2 + \theta rc^2\sigma_u^2)$, and the same steps as in Appendix B yield

$$b^*(c) = \frac{\delta^2\theta[L_l(c_{l,D} - c) + L_j\tau_j^2(c_{l,X} - c)]}{2\theta\gamma(\theta r\sigma_u^2 + \delta^2) - \delta^4c^2(L_l + L_j\tau_j^2)}.$$

The risk term in the denominator, $\theta r\sigma_u^2$, is now constant in c . Consider the domestic segment and write $A \equiv 2\theta\gamma(\theta r\sigma_u^2 + \delta^2)$, $N(c) = \delta^2\theta L_l(c_{l,D} - c)$ and $D(c) = A - \delta^4L_l c^2$. Differentiating,

$$\text{sign } b^{*'}(c) = \text{sign}[N'(c)D(c) - N(c)D'(c)] = \text{sign}[-A + \delta^4L_l(2c_{l,D}c - c^2)].$$

On $[0, c_{l,D}]$ the function $2c_{l,D}c - c^2$ is increasing and bounded by $c_{l,D}^2$, so the concavity condition for this specification, $A > \delta^4L_l c_{l,D}^2$ (the constant-variance analogue of Assumption A2(a)), implies $b^{*'}(c) < 0$ for every $c < c_{l,D}$. This also happens on the exporter segment if we replace L_l by $L_l + L_j\tau_j^2$ and $c_{l,D}$ by the corresponding weighted threshold. Thus, under constant noise variance, the optimal bonus is monotonically increasing in productivity, and

the hump of Prediction 1 disappears. Hence, the shape of the incentive schedule depends on whether compensation risk per unit of performance rises with productivity. The baseline specification implies this, as the managerial project puts the same absolute cost variance at stake in every firm, so relative to a productive firm's low cost base the risk is larger.

However, the two specifications are empirically distinguishable, and the data can help disentangle them. Constant per-unit noise implies monotone incentives, whereas the CEO data display the statistically significant inverse-U documented in Section 4, favoring the baseline specification.

D Equilibrium Cutoffs and Comparative Statics

For each country l , the zero-cutoff-profit condition (16) and the free-entry condition (17) jointly determine the domestic cutoff $c_{l,D}$ and the mass of entrants. Let $V(c_{l,D}; L_l, \tau_j) \equiv \int_{c_m}^{\hat{c}_l} E_u[\Pi(c)] dF(c)$ denote the expected value of entry. Since operating profits $\pi_{l,D} = \frac{L_l}{4\gamma}(c_{l,D} - c_e)^2$ and $\pi_{l,X} = \frac{L_j\tau_j^2}{4\gamma}(c_{l,X} - c_e)^2$ are increasing in the cutoff $c_{l,D}$ (a higher cutoff means weaker competition and larger price-cost margins), V is strictly increasing in $c_{l,D}$. Free entry implies $V(c_{l,D}; L_l, \tau_j) = f_E$.

A larger home market increases domestic profits at every realized cost, $\partial V/\partial L_l > 0$. Moreover, by the implicit function theorem,

$$\frac{\partial c_{l,D}}{\partial L_l} = -\frac{\partial V/\partial L_l}{\partial V/\partial c_{l,D}} < 0.$$

Thus, larger markets allow more entry, intensify competition, and lower the cutoff. Conversely, higher trade costs reduce export profits through $L_j\tau_j^2(c_{l,X} - c_e)^2$ with $c_{l,X} = c_{j,D}/\tau_j$. Thus, the net effect of a higher τ_j on the expected value of entry is negative, $\partial V/\partial \tau_j < 0$, and

$$\frac{\partial c_{l,D}}{\partial \tau_j} = -\frac{\partial V/\partial \tau_j}{\partial V/\partial c_{l,D}} > 0,$$

so higher trade barriers raise the cutoff and weaken selection. Note that these comparative statics hold the foreign cutoff fixed and thus do not account for the full two-country general-equilibrium effects. However, the counterfactuals of Section 5 solve the full two-country equilibrium system numerically.

E Labor Market Clearing and Welfare

Labor is the only factor and is used both in production and to cover the sunk entry cost. In country l , market clearing equates the labor endowment L_l to labor demanded by surviving firms for production across the domestic and export markets plus the labor absorbed by entry,²²

$$L_l = \underbrace{\int_{c_m}^{c_{l,D}} q_{l,D}(c_e) c_e dF N_E^l + \int_{c_m}^{c_{l,X}} \tau_j q_{l,X}(c_e) c_e dF N_E^l}_{\text{production}} + \underbrace{f_E N_E^l}_{\text{entry}},$$

which, together with the free-entry condition, pins down the mass of entrants N_E^l and the wage.

Welfare per capita is the sum of numeraire consumption and the surplus from the differentiated goods. Using the quadratic preferences underlying the demand system, the latter equals the term Z defined in the text,

$$W_l = w_l + \frac{1}{2} \left(\eta + \frac{\gamma}{N_l} \right)^{-1} (\alpha - \bar{p}_l)^2 + \frac{1}{2} \frac{N_l}{\gamma} \sigma_{p,l}^2,$$

where $\bar{p}_l$ is the average price and $\sigma_{p,l}^2$ the price dispersion in market l . A lower domestic cutoff $c_{l,D}$ increases consumer surplus and welfare as it lowers average realized costs and prices. Trade policy therefore affects welfare both directly, through prices and available varieties, and indirectly, through the managerial-incentive and selection channels that determine realized costs.

F Robustness: Perpetual-Inventory Capital Stock

The baseline productivity estimates of Section 3 measure the capital stock as deflated gross property, plant, and equipment. As a robustness check, we re-estimate total factor productivity by two-digit SIC industry, following the same ACF procedure as the baseline, but with a perpetual-inventory capital stock: $K_t = (1 - \delta_i)K_{t-1} + I_t$, where the initial stock is the firm's first observed real gross property, plant, and equipment, investment is real capital expenditures, and the firm-specific depreciation rate is estimated from reported property, plant, and equipment, capital expenditures, and depreciation. We then rebuild the 1993-base productivity index and re-estimate the two main specifications of Section 4.

²²As in the variety counts, participation is indexed by the initial draw: exact for domestic survival under A2(b); on the export margin the quantitative model uses realized-cost status.

Table 8 shows the relationship between productivity and incentives. The non-monotonic pattern of the baseline specification is preserved, i.e., the linear productivity term is positive and significant, the quadratic term negative, and across quintiles the effect on incentives peaks in the interior of the productivity distribution. Table 9 reports the specification regarding the effect of trade policy. The export-tariff interactions are negative and significant across the productivity distribution and survive the joint inclusion of both tariffs, while the import-tariff interactions largely lose significance, reproducing the heterogeneous effects of the baseline.

Table 8: Productivity and Incentives: Perpetual-Inventory Capital

	Quadratic Specification			Quintiles		
	(1)	(2)	(3)	(4)	(5)	(6)
$\log\omega_{j93}$	0.859** (0.367)	0.965*** (0.358)	0.444* (0.246)			
$\log\omega_{j93}^2$	-0.067 (0.046)	-0.073* (0.044)	-0.028 (0.030)			
q1 - $\log\omega_{j93}$				-0.279 (0.799)	-0.268 (0.768)	0.400 (0.363)
q2 - $\log\omega_{j93}$				1.407* (0.754)	1.272* (0.727)	0.981** (0.395)
q3 - $\log\omega_{j93}$				1.283* (0.741)	1.294* (0.724)	1.001*** (0.237)
q4 - $\log\omega_{j93}$				0.157 (0.547)	0.248 (0.502)	-0.040 (0.141)
Industry Controls	No	Yes	Yes	No	Yes	Yes
Firm and CEO Controls	No	No	Yes	No	No	Yes
R-Squared	0.89	0.89	0.91	0.89	0.89	0.91
N. Obs	6278	6278	5630	6278	6278	5630

Notes: Dependent variable is the log CEO delta. Productivity is estimated with a perpetual-inventory capital stock in place of deflated gross property, plant, and equipment. Columns (1)–(3) use a quadratic specification in baseline (1993) log productivity; Columns (4)–(6) replace it with productivity-quintile dummies (top quintile omitted). All columns include executive, industry, and year fixed effects; the “Controls” rows indicate industry and firm/CEO controls as in the baseline. Standard errors clustered by firm in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 9: Trade Liberalization and Incentives: Perpetual-Inventory Capital

	Tariffs			Interaction		
	(1)	(2)	(3)	(4)	(5)	(6)
Export Tariff	-0.920*** (0.051)		-0.742*** (0.055)	-0.018 (0.076)		-0.021 (0.076)
Import Tariff		-0.842*** (0.090)	-0.398*** (0.074)		-0.029 (0.076)	-0.069 (0.079)
q1 - $\log\omega_{j93}$				1.296** (0.589)	1.057* (0.555)	1.194** (0.581)
q2 - $\log\omega_{j93}$				1.391** (0.616)	0.902 (0.556)	1.369** (0.612)
q3 - $\log\omega_{j93}$				1.673*** (0.465)	1.360*** (0.463)	1.827*** (0.450)
q4 - $\log\omega_{j93}$				0.815** (0.397)	0.648 (0.396)	0.976** (0.416)
Export Tariff * q1 - $\log\omega_{j93}$				-0.393** (0.157)		-0.297** (0.144)
Export Tariff * q2 - $\log\omega_{j93}$				-0.584*** (0.172)		-0.525*** (0.173)
Export Tariff * q3 - $\log\omega_{j93}$				-0.405*** (0.135)		-0.308** (0.137)
Export Tariff * q4 - $\log\omega_{j93}$				-0.396*** (0.126)		-0.291** (0.126)
Import Tariff * q1 - $\log\omega_{j93}$					-0.324 (0.219)	-0.120 (0.227)
Import Tariff * q2 - $\log\omega_{j93}$					-0.503** (0.224)	-0.169 (0.214)
Import Tariff * q3 - $\log\omega_{j93}$					-0.464*** (0.153)	-0.266* (0.150)
Import Tariff * q4 - $\log\omega_{j93}$					-0.394*** (0.141)	-0.204 (0.137)
Controls	No	No	No	Yes	Yes	Yes
R-Squared	0.83	0.82	0.84	0.89	0.89	0.89
N. Obs	8520	8520	8520	5630	5630	5630

Notes: Dependent variable is the log CEO delta. Productivity quintiles are based on the perpetual-inventory capital stock. Columns (1)–(3) report tariff levels; Columns (4)–(6) interact tariffs with baseline-productivity quintiles (top quintile omitted). All columns include executive and industry fixed effects and the controls indicated. Standard errors clustered by firm in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

G Average Tariffs by Industry

Table 10 reports average U.S. import and export tariffs by two-digit SIC industry in 1993 and 2007. Relatively labor-intensive sectors such as apparel and leather (SIC 23 and 31) face higher protection than capital-intensive sectors such as paper, chemicals, and petroleum

(SIC 26–29). During the period of analysis, tariffs fell nearly across the board (except for the export tariff on food products), but the pace of liberalization differed substantially across industries.

Table 10: Average Ad-Valorem Tariffs by Two-Digit SIC Industry, United States

	1993	2007	1993	2007
20	3.34	2.08	11.62	11.98
22	10.59	5.68	11.57	3.40
23	12.44	8.31	13.17	5.50
24	0.88	0.40	3.93	1.74
25	2.95	0.22	3.75	0.91
26	0.64	0.14	5.32	0.71
27	0.97	0.07	1.46	0.17
28	4.13	1.49	6.62	2.49
29	0.77	0.34	2.46	0.61
30	7.23	4.07	8.01	2.37
31	10.35	7.97	13.35	5.90
32	6.00	2.94	5.82	2.06
33	3.33	0.46	5.02	1.28
34	3.47	1.05	6.37	1.48
35	2.36	0.51	4.82	1.45
36	3.85	0.65	6.23	1.71
37	2.54	0.67	6.53	2.06
38	4.83	1.12	5.60	1.69
39	5.24	2.22	6.62	2.06
Total	4.52	2.13	6.75	2.61

Notes: Average firm-level ad-valorem tariffs by two-digit SIC industry, weighted by 1993 trade shares. The first two columns report import tariffs in 1993 and 2007; the last two report export tariffs in 1993 and 2007. The final row reports the all-industry average.

H Productivity and Incentives: OLS and Lagged Productivity

Table 11 re-estimates the productivity-incentive relationship of equation (20) using ordinary-least-squares total factor productivity in place of the ACF measure, both in quadratic form (Columns 1–3) and in quintiles (Columns 4–6), and using lagged ($t - 2$) productivity (Columns 7–9). The non-monotonic pattern is preserved throughout: the linear productivity term is positive and the quadratic term negative and significant, and across quintiles the slope is steepest at the bottom and flattens at the top, as in the baseline.

Table 11: Productivity and Incentives: OLS and Lagged Productivity

	Quadratic Specification OLS			Quintiles OLS			Lagged Productivity - t-2		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
$\log \omega_{j93}$	14.349*** (2.497)	15.700*** (2.328)	6.919** (3.245)						
$\log \omega_{j93}^2$	-1.533*** (0.266)	-1.659*** (0.254)	-0.749** (0.364)						
q1 - $\log \omega_{j93}$				1.550*** (0.575)	1.365** (0.576)	1.218*** (0.431)			
q2 - $\log \omega_{j93}$				2.016*** (0.401)	1.934*** (0.389)	1.349*** (0.428)			
q3 - $\log \omega_{j93}$				1.904*** (0.525)	1.948*** (0.525)	1.608*** (0.474)			
q4 - $\log \omega_{j93}$				0.680** (0.287)	0.737*** (0.273)	0.533* (0.321)			
$\log \omega_{jt-2}$							1.137*** (0.256)	1.220*** (0.255)	1.012*** (0.227)
$\log \omega_{jt-2}^2$							-0.048*** (0.012)	-0.052*** (0.012)	-0.046*** (0.010)
Controls	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes
R-Squared	0.89	0.89	0.91	0.89	0.89	0.91	0.90	0.90	0.91
N. Obs	6278	6278	5630	6278	6278	5630	7310	7310	7168

Notes: Dependent variable is the log CEO delta. Columns (1)–(3) use a quadratic specification in OLS total factor productivity; Columns (4)–(6) replace it with OLS productivity quintiles; Columns (7)–(9) use two-year-lagged productivity. All columns include the controls (the “Controls” row) and fixed effects described in Section 4. Standard errors in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

I Trade Liberalization and Incentives: Year Fixed Effects

Table 12 re-estimates the six columns of the trade-liberalization specification of equation (21) adding year fixed effects to every column. With the introduction of year fixed effects, the aggregate coefficients in Columns 1–3 are no longer identified from the common time variation. However, the heterogeneity results are robust to this inclusion. In particular, the export-tariff interactions remain negative and significant with the same declining pattern across quintiles (−0.59 in the lowest quintile, −0.26 and −0.27 in the middle quintiles, and an insignificant −0.14 in the fourth), while the import-tariff interactions remain insignificant throughout.

Table 12: Trade Liberalization and Incentives: Year Fixed Effects

	Tariffs			Interaction		
	(1)	(2)	(3)	(4)	(5)	(6)
Export Tariff	-0.098 (0.064)		-0.106 (0.067)	0.255*** (0.074)		0.258*** (0.075)
Import Tariff		0.001 (0.057)	0.028 (0.060)		0.121 (0.101)	0.020 (0.082)
q1 - $\log\omega_{j93}$				2.376*** (0.586)	1.781*** (0.605)	2.246*** (0.607)
q2 - $\log\omega_{j93}$				1.769*** (0.415)	1.539*** (0.421)	1.803*** (0.441)
q3 - $\log\omega_{j93}$				1.856*** (0.401)	1.560*** (0.396)	1.864*** (0.431)
q4 - $\log\omega_{j93}$				0.654* (0.356)	0.503 (0.368)	0.691* (0.400)
Export Tariff * q1 - $\log\omega_{j93}$				-0.457** (0.190)		-0.594*** (0.201)
Export Tariff * q2 - $\log\omega_{j93}$				-0.295** (0.125)		-0.259** (0.118)
Export Tariff * q3 - $\log\omega_{j93}$				-0.236** (0.116)		-0.274** (0.113)
Export Tariff * q4 - $\log\omega_{j93}$				-0.165 (0.113)		-0.142 (0.115)
Import Tariff * q1 - $\log\omega_{j93}$					-0.096 (0.250)	0.265 (0.268)
Import Tariff * q2 - $\log\omega_{j93}$					-0.238 (0.159)	-0.105 (0.142)
Import Tariff * q3 - $\log\omega_{j93}$					-0.048 (0.162)	0.106 (0.152)
Import Tariff * q4 - $\log\omega_{j93}$					-0.119 (0.137)	-0.054 (0.132)
Controls	No	No	No	Yes	Yes	Yes
R-Squared	0.87	0.87	0.87	0.91	0.91	0.91
N. Obs	8520	8520	8520	5630	5630	5630

Notes: Dependent variable is the log CEO delta. All columns replicate Table 3 with year fixed effects added: Columns (1)–(3) report tariff levels; Columns (4)–(6) interact tariffs with baseline (1993) productivity quintiles (top quintile omitted). All columns include executive, industry (2-digit SIC), and year fixed effects; Columns (4)–(6) add the controls indicated. Standard errors clustered by firm in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

J Trade Liberalization and Incentives: Robustness

Table 13 re-estimates equation (21) using a quadratic tariff-productivity interaction (Columns 1–3) and OLS total factor productivity quintiles in place of the baseline ACF quintiles (Columns 4–6). The pattern of heterogeneity remains. In the quadratic specification the two

export-tariff interaction terms are jointly significant ($p = 0.004$ in the export-only column, $p = 0.001$ with both tariffs included). Moreover, the implied export-tariff response declines monotonically across the productivity distribution, from -0.54 at the 10th percentile of baseline productivity to -0.14 at the 90th. By contrast, the import-tariff interactions are jointly insignificant ($p = 0.38$ and $p = 0.71$). The quintile columns show the same pattern.

Table 13: Trade Liberalization and Incentives: Robustness

	Quadratic Specification			Quantiles OLS		
	(1)	(2)	(3)	(4)	(5)	(6)
Export Tariff	-4.668 (9.158)		-10.687 (8.081)	-0.035 (0.076)		-0.028 (0.078)
Import Tariff		6.560 (14.229)	12.089 (15.004)		-0.043 (0.074)	-0.075 (0.078)
$\log\omega_{j93}$	4.487 (5.670)	4.643 (6.114)	6.207 (6.379)			
$\log\omega_{j93}^2$	-0.263 (0.295)	-0.254 (0.320)	-0.349 (0.334)			
Export Tariff * $\log\omega_{j93}$	0.516 (1.775)		1.716 (1.559)			
Export Tariff * $\log\omega_{j93}^2$	-0.008 (0.086)		-0.068 (0.075)			
Import Tariff * $\log\omega_{j93}$		-1.579 (2.804)	-2.436 (2.953)			
Import Tariff * $\log\omega_{j93}^2$		0.089 (0.139)	0.121 (0.146)			
q1 - $\log\omega_{j93}$				1.335* (0.730)	1.014 (0.690)	1.285* (0.731)
q2 - $\log\omega_{j93}$				1.714** (0.676)	1.399** (0.671)	1.846*** (0.685)
q3 - $\log\omega_{j93}$				2.235*** (0.786)	1.868** (0.746)	2.341*** (0.794)
q4 - $\log\omega_{j93}$				1.077* (0.601)	0.937* (0.563)	1.159** (0.582)
Export Tariff * q1 - $\log\omega_{j93}$				-0.512** (0.232)		-0.531** (0.235)
Export Tariff * q2 - $\log\omega_{j93}$				-0.469*** (0.124)		-0.367*** (0.122)
Export Tariff * q3 - $\log\omega_{j93}$				-0.455*** (0.125)		-0.333*** (0.128)
Export Tariff * q4 - $\log\omega_{j93}$				-0.267** (0.121)		-0.179 (0.116)
Import Tariff * q1 - $\log\omega_{j93}$					-0.249 (0.287)	0.069 (0.295)
Import Tariff * q2 - $\log\omega_{j93}$					-0.439*** (0.152)	-0.211 (0.154)
Import Tariff * q3 - $\log\omega_{j93}$					-0.583*** (0.135)	-0.339** (0.140)
Import Tariff * q4 - $\log\omega_{j93}$					-0.279** (0.142)	-0.161 (0.136)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
R-Squared	0.89	0.89	0.89	0.89	0.89	0.89
N. Obs	5630	5630	5630	5630	5630	5630

Notes: Robustness of the export- and import-tariff specification of Section 4. Columns (1)–(3) use a quadratic tariff-productivity interaction; Columns (4)–(6) use OLS total factor productivity quintiles. All columns include the controls (the “Controls” row) and fixed effects. Standard errors in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

K Robustness: Sensitivity to Foreign-Market Size

The baseline calibration sets the foreign-to-home market-size ratio to 3, matching the U.S. share of roughly one quarter of world GDP. However, given that the size of the foreign market scales the export channel through which managerial incentives respond to trade policy, we assess how the calibration and counterfactuals change under alternative values of this ratio. To do this, we re-estimate the seven structural parameters at $L_f \in \{1, 2, 4\}$ and recompute the counterfactuals at each value. Table 14 reports the results.

Table 14: Robustness: Re-estimation at Alternative Foreign-Market Sizes

L_f	SMM loss	δ	θ	Export $\tau_f+0.18$		No bonus
				fixed cutoffs	full eq.	full eq.
1	0.362	1.71	9.50	-1.35	-6.00	-11.29
2	0.277	1.39	8.14	-0.16	-6.16	-13.15
3 (baseline)	0.185	1.71	9.53	-0.07	-7.61	-13.98
4	0.168	1.78	8.67	-0.21	-8.66	-14.98

Notes: Each row re-estimates the seven structural parameters at a fixed foreign-market size L_f (home $L = 1$), holding the specification otherwise unchanged (the same seven moments, the effort-contribution anchor, and $\gamma = 0.25$ fixed). $L_f = 3$ is the baseline (US/rest-of-world GDP ratio). The last three columns report the change in productivity ($1/\bar{c}_e$, in %): the fixed-cutoff and full general-equilibrium effect of an increase of 0.18 in the home export barrier τ_h , and the full-GE effect of removing managerial bonuses. The import tariff τ_h is incentive-neutral at every L_f (within-firm effect = 0) and is omitted for brevity.

Three patterns stand out. First, the SMM criterion declines with L_f (0.36 at the symmetric case $L_f = 1$, 0.19 at the baseline, 0.17 at $L_f = 4$), so the baseline is disciplined by the observable GDP share and fits far better than symmetry. Second, the qualitative conclusions are stable across the new specifications. As the model proposes, an increase in the export barrier lowers productivity through selection at every L_f . Moreover, the general-equilibrium effect is larger at larger foreign-market sizes (from -6.0% at $L_f = 1$ to -8.7% at $L_f = 4$). Third, the fixed-cutoff productivity response to the export barrier is small at every level of L_f . This is because the incentive damage appears in pay and output rather than in the fixed-cutoff productivity index.

L Robustness: Sensitivity to Product Differentiation

The baseline fixes the product-differentiation parameter at $\gamma = 0.25$ because it is weakly identified by our moments (Section 5). To assess whether this choice matters, we re-estimate

the model at $\gamma = 0.15$ and $\gamma = 0.35$ and recompute the counterfactuals at each estimate. Table 15 reports the results and confirms they are robust to the choice of γ .

In particular, the minimized criterion is nearly flat across γ , the anchored effort contribution binds at 6.0% in every column, and the counterfactuals barely move. The general-equilibrium productivity loss from an increase of 0.18 in the export barrier τ_f is between 7.6% and 7.8% across the range of γ , the fixed-cutoff effect remains near zero in all cases, and removing bonuses lowers productivity by 13.5 to 14.2 percent in each scenario. The general-equilibrium output response varies somewhat more (from -4.0% to -6.1% , against -5.2% at the baseline), but the qualitative conclusions, and in particular the importance of the export margin, remain.

Table 15: Robustness: Re-estimation at Alternative Values of γ

γ	SMM loss	δ	θ	Effort contr. (%)	Export $\tau_f+0.18$		
					prod. fixed cutoffs	prod. full eq.	output full eq.
0.15	0.187	1.57	9.01	6.0	-0.07	-7.57	-6.05
0.25 (baseline)	0.185	1.71	9.53	6.0	-0.07	-7.61	-5.15
0.35	0.170	1.53	10.04	6.0	+0.42	-7.84	-4.03

Notes: Each row re-estimates the seven structural parameters at a fixed value of the product-differentiation parameter γ , holding the specification otherwise unchanged (the same seven moments, the effort-contribution anchor, and $L_f = 3$). $\gamma = 0.25$ is the baseline. The effort contribution is the partial-equilibrium productivity loss from switching performance pay off at fixed cutoffs (anchored at 6%). The last three columns report responses to an increase of 0.18 in the home export barrier τ_h : the fixed-cutoff and general-equilibrium change in productivity ($1/\bar{c}_e$) and the general-equilibrium change in output, all in percent.

M Identification of the Moral-Hazard Block

The seven shape moments identify only combinations of the moral-hazard parameters (δ, θ, σ) . To document this, we refine the grid estimate by local optimization (trust-region least squares, multistart, fixed simulation draws) under progressively wider parameter bounds, and separately re-estimate the six remaining parameters at fixed values of θ . Two findings emerge. First, whenever the bounds are widened, the optimizer drives θ to the upper bound offered and δ toward its lower bound, with the criterion improving only marginally while the implied managerial effort falls: the criterion is nearly flat along this direction. Second, fixing θ and re-estimating the rest reproduces the same pattern in miniature, as Table 16 shows: the criterion varies little and non-monotonically in θ over a wide range, while the implied contribution of effort to within-firm productivity varies severalfold and δ sits at its bound in every column.

Table 16: The Scale of the Moral-Hazard Block Is Not Identified by the Shape Moments

θ (fixed)	6.5	7.5	8.6	10	12
SMM criterion (7 shape moments)	0.118	0.139	0.114	0.107	0.108
Effort contribution to productivity (%)	1.2	1.2	0.8	0.5	0.5
$\hat{\delta}$	1.20	1.20	1.20	1.20	1.20

Notes: Each column re-estimates the six remaining parameters on the seven shape moments at the indicated fixed θ (multistart local optimization, fixed simulation draws). The effort contribution is the partial-equilibrium productivity loss from switching performance pay off at fixed cutoffs. $\hat{\delta} = 1.20$ is the lower bound of the parameter space in every column.

Table 17 reports the parameter estimates with their standard errors. The noise parameter σ and the export cost τ_f are precisely estimated, the Pareto shape less sharply, and the import cost τ_h imprecisely, consistent with its indirect, general-equilibrium identification. The moral-hazard parameters δ and θ are individually imprecise; however, their combination, which is the one that matters for the counterfactuals, is tightly determined.

Table 17: SMM Parameter Estimates with Standard Errors

Parameter	Estimate	Std. err.
Entry cost (f_E)	2.77×10^{-2}	(0.058)
Marginal product of effort (δ)	1.715	(2.257)
Noise std. dev. (σ)	0.209	(0.027)
Pareto shape (κ)	2.275	(0.977)
Effort disutility (θ)	9.528	(47.244)
Product differentiation (γ , fixed)	0.250	—
Trade cost, home $\rightarrow$ foreign (τ_f)	1.287	(0.193)
Trade cost, foreign $\rightarrow$ home (τ_h)	1.864	(1.385)

Notes: Delta-method sandwich standard errors combining the numerical moment Jacobian, a firm-level bootstrap of the empirical moments, and the simulation-noise covariance, conditional on the effort-contribution anchor (a normalization). δ and θ are individually imprecise along the near-flat direction documented in Online Appendix M; the anchored effort contribution itself is tightly determined, with a 95 percent confidence interval of 5.9 to 6.1 percent.

N Robustness: The Effort-Contribution Anchor

The baseline estimation anchors the effort contribution at 6% following Bloom and Van Reenen (2007). To assess how much the quantitative conclusions depend on this normalization, we re-estimate the model with the anchor at the ends of the range proposed in the literature: 3% and 8%. Table 18 shows the results.

The parameters move as Online Appendix M predicts: a smaller anchored contribution

lowers δ and raises θ , while the trade costs are essentially unchanged. This is reassuring, as it confirms that the trade environment is identified independently of the anchor. Moreover, the minimized criterion is nearly as good across the whole range (0.203, 0.185, 0.190). While the quantitative magnitudes scale with the anchor, as expected, the conclusions do not change qualitatively. For instance, the no-bonus economy is 9 to 16 percent less productive, the bilateral trade war reduces output by 3.6 to 5.8 percent with performance pay but by only about 1 percent without it, and the share of the output cost that comes from the incentive channel ranges from roughly two thirds at the most conservative anchor to four fifths at the baseline and above. That is, within this range proposed by the literature, a model without managerial incentives understates the output cost of the trade war by a factor of three to five.

Table 18: Robustness: Alternative Effort-Contribution Anchors

Anchor	SMM loss	δ	θ	Effort contr. (%)	No-bonus prod. (%)	Bilateral trade war, output		
						with pay	without pay	share (%)
3%	0.203	1.47	13.72	3.0	-9.4	-3.6	-1.3	64
6% (baseline)	0.185	1.71	9.53	6.0	-14.0	-5.0	-1.0	81
8%	0.190	1.74	8.53	8.0	-16.3	-5.8	-1.1	80

Notes: Each row re-estimates the seven structural parameters with the effort-contribution anchor set to the indicated value. 6% is the baseline; 3% and 8% are the ends of the range in Bloom and Van Reenen (2007). The effort contribution is the fixed-cutoff productivity loss from removing the managerial project. No-bonus prod. is the full-equilibrium productivity gap between the estimated economy and the no-bonus economy. The last three columns show the change in output following a bilateral trade war in an economy with and without performance pay, each relative to its own baseline. The share refers to the difference in those percentages.